Respect the Unstable

The practical, physical (and sometimes dangerous) consequences of control must be respected, and the underlying principles must be clearly and well taught.

By Gunter Stein

Feedback control systems are all around us in modern technological life. They are at work in our homes, our cars, our factories, our transportation systems, our defense systems—everywhere we look. Certainly, one of the great achievements of the international controls research community is that the design principles for these systems are well developed and broadly understood by control engineers, so that the systems operate productively and safely in so many applications.

In this article, I want to talk about two trends that threaten to undermine this achievement. My objective is to heighten our awareness of these trends and hopefully bring about an appropriate response to them.

The first trend has to do with the applications themselves. Among the abundance of control systems operating today are increasing numbers of dangerous ones. Society trusts our technology. We are permitted to do things with automatic controls that cannot be done manually and that, if done improperly, can have dire consequences for property, the environment, and human life. Most, but not all, of these dangerous applications involve open-loop unstable plants with divergence rates violent enough to elude manual control. This characterization motivates the title of the article, and I will describe specific examples of such applications.

The second trend has been evident at our conferences, and certainly in our journals, over the years. This trend is the increasing worship of abstract mathematical results in control at the expense of more specific examinations of their practical, physical consequences. I will provide examples of this trend as well.

Gunter Stein's Bode Lecture

An understanding of fundamental limitations is an essential element in all engineering. Shannon's early results on channel capacity have always had center court in signal processing. Strangely, the early results of Bode were not accorded the same attention in control. It was therefore highly appropriate that the IEEE Control Systems Society created the Bode Lecture Award, an honor which also came with the duty of delivering a lecture. Gunter Stein gave the first Hendrik W. Bode Lecture at the IEEE Conference on Decision and Control in Tampa, Florida, in December 1989. In his lecture he focused on Bode's important observation that there are fundamental limitations on the achievable sensitivity function expressed by Bode's integral. Gunter has a unique position in the controls community because he combines the insight derived from a large number of industrial applications at Honeywell with long experience as an influential adjunct professor at the Massachusetts Institute of Technology from 1977 to 1996. In his lecture, Gunter also emphasized the importance of the interaction between instability and saturating actuators and the consequences of the fact that control is becoming increasingly mission critical.

After more than 13 years I still remember Gunter's superb lecture. I also remember comments from young control scientists who had been brought up on state-space theory who said: "I believed that controllability and observability were the only things that mattered." At Lund University we made Gunter's lecture a key part of all courses in control system design. Gunter was brought into the classroom via videotapes published by the IEEE Control Systems Society and the written lecture notes. It was a real drawback that the lecture was not available in more archival form. I am therefore delighted that *IEEE Control Systems Magazine* is publishing this article. I sincerely hope that this will be followed by a DVD version of the videotape. The lecture is like really good wine; it ages superbly.

—Karl J Åström, Professor Emeritus
Lund University, Lund, Sweden

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Respect the Unstable

必须重视控制所产生的实际物理后果（有时甚至是危险的后果），并且必须清晰、透彻地传授其基本原理。

作者：Gunter Stein

Feedback

在现代技术生活中，控制系统无处不在。它们在我们的家庭、汽车、工厂、交通运输系统、国防系统中发挥作用——放眼望去，处处可见。当然，国际控制研究界的一项重大成就是，这些系统的设计原理已得到充分发展，并为控制工程师广泛理解，因此这些系统能够在众多应用中高效、安全地运行。

在本文中，我想讨论两种可能削弱这一成就的趋势。我的目的是提高我们对这些趋势的认识，并希望促使我们作出适当的应对。

第一种趋势与应用本身有关。在当今运行的大量控制系统中，危险系统的数量正不断增加。社会信任我们的技术。我们获准使用自动控制来完成人工无法完成的操作，而这些操作一旦执行不当，就可能对财产、环境和人的生命造成严重后果。这些危险应用大多涉及开环不稳定的被控对象，其发散速率剧烈到人工控制无法应对，但并非所有危险应用都如此。这一特征促成了本文的标题，我将介绍此类应用的具体实例。

多年来，第二种趋势在我们的会议上已显而易见，在我们的期刊中更是如此。这种趋势是，对控制领域抽象数学结果的崇拜日益加深，却忽视了对其实际物理后果更为具体的考察。我也将举例说明这一趋势。

对 Gunter Stein 的 Bode 讲座

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基本局限性的理解是所有工程领域不可或缺的要素。Shannon 关于信道容量的早期成果始终在信号处理中占据核心地位。奇怪的是，Bode 的早期成果在控制领域却没有受到同等重视。因此，IEEE 控制系统学会设立 Bode 讲座奖是十分恰当的，这一荣誉也附带了发表演讲的义务。1989 年 12 月，Gunter Stein 在佛罗里达州坦帕举行的 IEEE 决策与控制会议上作了首场 Hendrik W. Bode 讲座。在演讲中，他重点阐述了 Bode 的重要发现：可实现的灵敏度函数存在由 Bode 积分所表达的基本局限性。Gunter 在控制界具有独特地位，因为他将从 Honeywell 的大量工业应用中获得的洞见，与 1977 年至 1996 年在麻省理工学院担任颇具影响力的兼职教授的长期经验相结合。在演讲中，Gunter 还强调了不稳定性与执行器饱和之间相互作用的重要性，以及控制对任务成败日益关键这一事实所带来的后果。

13年多过去了，我仍然记得 Gunter 那场精彩的讲座。我也记得一些以状态空间理论为基础接受培养的年轻控制科学家的评论，他们说："我原以为，可控性和可观性是唯一重要的东西。"在隆德大学，我们将 Gunter 的讲座列为所有控制系统设计课程的重要组成部分。通过 IEEE 控制系统学会出版的录像带和书面讲义，Gunter 得以走进课堂。这场讲座未能以更便于长期保存和查阅的形式提供，实在是一大遗憾。因此，我很高兴*IEEE* 控制系统杂志刊登这篇文章。我衷心希望，随后还能推出该录像带的 DVD 版本。这场讲座犹如佳酿，历久弥醇。

——Karl J Åström，瑞典隆德，
隆德大学荣休教授

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Together, these trends threaten to undermine our good standing in society as masters of a technology that can be trusted.

The Punch Line

In slightly dramatized form, the message of this article can be summarized by drawing some contrasts. Consider the following mathematical statement:

Theorem: Given plant $g(s)$, compensator $k(s)$, and

$$gk(s) = n(s)/d(s); \quad s(s) = 1/[1 + gk(s)]$$
$$Z = \{z \mid n(z) = 0, \mathrm{Re}(z) \geq 0\}$$
$$P = \{p \mid d(p) = 0, \mathrm{Re}(p) \geq 0\}.$$

Then $s(s)$ is stable if and only if

$$s(s) < \infty \quad \forall\, \mathrm{Re}(s) \geq 0$$
$$s(z) = 1 \quad \forall\, z \in Z$$
$$s(p) = 0 \quad \forall\, p \in P.$$

In words, this theorem states that the sensitivity function of a feedback system must not only be finite in the right-half plane, but it must pass through certain interpolation points corresponding to right-half-plane singularities of the loop. Most of us recognize this immediately as an elegant and compact description of control system constraints imposed by unstable, nonminimum-phase systems. It was formally developed in the context of parametrizing all stabilizing controllers, and it was popularized in the 1980s as part of the interpolation-theoretic approach to H_∞ optimization. (Of course, it was understood as stated above for single-input, single-output (SISO) systems as far back as the 1950s. Some historical notes on this theorem can be found in [1].)

Unfortunately, we are not as quick to recognize that this mathematical description includes some very dangerous systems. For example, the theorem applies to the JAS-39 airplane (the SAAB Gripen), which crashed on landing in 1989 in one of its first test flights. Figure 1 shows a video frame from the crash. Fortunately, the pilot survived, but the airplane was lost and its development program substantially delayed.

The theorem also applies to the Chernobyl nuclear plant, shown in Figure 2 as it appeared shortly after its accident in 1986. We are all familiar with the consequences of that accident—hundreds of people dead, hundreds of thousands evacuated, and hundreds of millions of dollars in cleanup costs.

These and other examples dramatize the contrast between elegant mathematical statements and the real physical systems that they purport to describe. I have selected these two examples because both catastrophes involve explicit, traceable responsibilities of control systems. We will take a closer look at those responsibilities later.

Basic Facts About Unstable Plants

- Unstable systems are fundamentally, and quantifiably, more difficult to control than stable ones.
- Controllers for unstable systems are operationally critical.
- Closed-loop systems with unstable components are only locally stable.

My point is this: As society permits control engineers to operate more such dangerous systems, we who teach those engineers and fashion their tools cannot hide from responsibility under a cloak of mathematics. We dare not instill the notion that mathematical rigor is the only goal to strive for in control. We must also instill respect for the practical, physical consequences of control, and we must make certain that its underlying principles are taught clearly and well.

I want to explore this point by reviewing some basic facts about unstable plants. Again, we will consider *unstable* to be synonymous with *dangerous,* even though this is not all inclusive. I want to review these facts:

- unstable systems are fundamentally, and quantifiably, more difficult to control than stable ones
- controllers for unstable systems are operationally critical
- closed-loop systems with unstable components are only locally stable.

These facts should be well known to all of us, but as we will see, they are not always taken to heart.



Figure 1. *Gripen JAS39 prototype accident on 2 February 1989. The pilot received only minor injuries.*

这些趋势共同威胁着我们在社会中的良好声誉，即我们作为值得信赖的技术的掌握者所享有的声誉。

点睛之笔

稍加戏剧化处理，本文的主旨可以通过一些对比来概括。请看以下数学陈述：

定理：给定被控对象 $g\,s(\,)$、补偿器 $k\,s(\,)$，以及

$$gk(s) = n(s)/d(s); \quad s(s) = 1/[1 + gk(s)]$$
$$Z = \{z \mid n(z) = 0, \mathrm{Re}(z) \geq 0\}$$
$$P = \{p \mid d(p) = 0, \mathrm{Re}(p) \geq 0\}.$$

则 $s\,s(\,)$ 稳定，当且仅当

$$s(s) < \infty \quad \forall\, \mathrm{Re}(s) \geq 0$$
$$s(z) = 1 \quad \forall\, z \in Z$$
$$s(p) = 0 \quad \forall\, p \in P.$$

用文字表述，这一定理指出，反馈系统的灵敏度函数不仅必须在右半平面内有限，还必须通过某些插值点，这些插值点对应于环路在右半平面内的奇点。我们大多数人一眼就能看出，这是对不稳定、非最小相位系统所施加的控制系统约束的优雅而简洁的描述。该定理是在对所有镇定控制器进行参数化的背景下正式建立的，并于 1980s 作为基于插值理论的 H_∞ 优化方法的一部分得到推广。（当然，早在 1950s，人们就已经对单输入单输出（SISO）系统有了上述认识。关于这一定理的一些历史说明可见 in [1].）

遗憾的是，我们并未同样迅速地意识到，这一数学描述涵盖了一些非常危险的系统。例如，该定理适用于JAS-39飞机（SAAB鹰狮），这架飞机于1989年在最初几次试飞中的一次着陆时坠毁。图1展示了事故视频中的一帧画面。幸运的是，飞行员幸存，但飞机报废，其研制计划也遭到严重延误。

该定理也适用于切尔诺贝利核电站，图2展示了它在1986年事故发生后不久的景象。我们都熟知那场事故的后果——数百人死亡，数十万人被疏散，清理费用达数亿美元。

这些例子以及其他例子鲜明地展示了优雅的数学表述与其声称描述的真实物理系统之间的反差。我选择这两个例子，是因为这两起灾难都涉及控制系统明确的、可追溯的责任。稍后我们将更仔细地审视这些责任。

关于不稳定被控对象的基本事实

•不稳定系统在根本上比稳定系统更难控制，而且这种难度差异可以量化。• 不稳定系统的控制器对系统运行至关重要。• 含有不稳定部件的闭环系统仅具有局部稳定性。

我想强调的是：随着社会允许控制工程师操作更多此类危险系统，我们这些培养工程师并为他们打造工具的人，不能披着数学的外衣逃避责任。我们绝不能灌输这样的观念：数学严谨性是控制领域唯一值得追求的目标。我们还必须培养对控制所带来的实际物理后果的敬畏，并确保其基本原理得到清晰、充分的讲授。

我想通过回顾关于不稳定被控对象的一些基本事实来探讨这一点。再次强调，我们将把不稳定视为危险的同义词，尽管这并不能涵盖所有情况。我想回顾以下事实：

• 不稳定系统从根本上比稳定系统更难控制，而且这种差异可以量化• 不稳定系统的控制器对系统运行至关重要• 含有不稳定部件的闭环系统仅具有局部稳定性。

这些事实应该是我们所有人都熟知的，但正如我们将看到的，人们并不总是将其铭记于心。



图 1. *Gripen JAS39* 原型机于 *1989* 年 *2* 月 *2* 日发生事故。飞行员仅受轻伤。

Bode Integrals

The first integral applies to stable plants and the second to unstable plants. They are valid for every stabilizing controller, assuming only that both plant and controller have finite bandwidths. In words, the integrals state that the log of the magnitude of sensitivity function of a SISO feedback system, integrated over frequency, is constant. The constant is zero for stable plants, and it is positive for unstable ones. It becomes larger as the number of unstable poles increases and/or as the poles move farther into the right-half plane. (Technically, we must count all unstable poles here, including those in the compensator, if any.)

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = 0$$

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = \pi \sum_{p\in P} \mathrm{Re}(p)$$

Fact 1: Fundamental, Quantifiable Control Difficulty

One of the best-known examples of instability is the inverted pendulum, or the "broomstick balancing problem." Those of us who learned our craft in the 1950s and 1960s know this problem well, either through the little cart experiments that appeared during those years in various university labs around the world or through textbook examples and simulations. Recall that this problem was motivated by the space race. Control engineers had to learn how to balance rockets on top of their plumes on their way to earth orbit. It is still an important problem today, in the post-Challenger era, as modern boosters, now built in several countries around the world, become more difficult to balance.

I bring up the inverted pendulum because it nicely illustrates our first fact. The illustration is this: I can obviously balance an ordinary stable pendulum without difficulty. I can also easily balance a long inverted pendulum, even in front of a room full of people. However, I find it more difficult to balance a shorter inverted pendulum, and I find it impossible to balance a very short one. You have probably tried this yourself. The exact lengths you can balance might be different, but the trend will be the same.

Certainly the theorem just cited has an explanation for this illustration hidden in it somewhere, and we could try to extract that explanation using all the modern machinery at our disposal. Perhaps we could compute some minimum H_∞ norms or even some minimum structured singular values achievable by human controllers. Of course, in our calculations, we would need to pay due attention to the inherent handicaps of such controllers, such as reaction time, neuromuscular lags, limb inertias, and many other uncertainties covering the fact that humans, and most everything else in the physical world, are not finite dimensional, linear, and time invariant.

The Bode Integrals

I will try to explain these observations in the frequency domain the way Hendrik Bode might have done it. It turns out that such an explanation is insightful, plain, and clear and is therefore preferable to many modern ones.

First, note that the difference between balancing long sticks and short sticks has to do with the location of the unstable mode. This is obvious from the linearized equations of motion of the stick. Under the simplifying assumption that all mass is concentrated at the end of the stick, these equations show an unstable pole at $\sqrt{g/L}$, where g denotes the acceleration of gravity and L denotes length. Divergence becomes more rapid as L decreases.

A frequency domain quantification of control difficulty in the face of such changing instabilities is captured by the Bode integrals (see sidebar). At the risk of sounding dogmatic, I believe that every control theoretician and every control engineer should know these integrals and understand their meaning. Unfortunately, we have not always taught them well.

My own background illustrates this last observation. You can judge for yourself, using your own background. During my entire control education as an undergraduate and graduate student in the 1960s, I ran across only the first integral, only



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Figure 2. *Chernobyl nuclear power plant shortly after the accident on 26 April 1986.*

波德积分

第一个积分适用于稳定的被控对象，第二个积分适用于不稳定的被控对象。它们对所有镇定控制器都成立，仅假设被控对象和控制器均具有有限带宽。用文字表述，这些积分说明：SISO 反馈系统的灵敏度函数幅值的对数在频率上的积分为常数。对于稳定的被控对象，该常数为零；对于不稳定的被控对象，该常数为正。随着不稳定极点数量的增加和/或极点向右半平面更深处移动，该常数会增大。（严格地说，这里必须计入所有不稳定极点，包括补偿器中的不稳定极点，如果有的话。）

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = 0$$

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = \pi \sum_{p\in P} \mathrm{Re}(p)$$

事实 1：基本的、可量化的控制难度

最广为人知的不稳定性例子之一是倒立摆，或称为“扫帚杆平衡问题”。我们这些在1950年代和1960年代学习本专业的人，对这个问题都很熟悉，无论是通过那些年世界各地大学实验室中出现的小车实验，还是通过教科书中的例子和仿真。回想一下，这个问题的提出源于航天竞赛。控制工程师必须学会如何在火箭飞往地球轨道的途中，使其在尾焰上保持平衡。在挑战者号事故之后的时代，这至今仍是一个重要问题，因为如今由世界上多个国家制造的现代助推器越来越难以保持平衡。

我提到倒立摆，是因为它很好地说明了我们的第一个事实。具体来说：显然，我可以毫不费力地使一个普通的稳定摆保持平衡。我也可以轻松地使一个长倒立摆保持平衡，即使面对满屋子的人也是如此。然而，我发现，使较短的倒立摆保持平衡更加困难，而使非常短的倒立摆保持平衡则是不可能的。你很可能也亲自尝试过。你能保持平衡的具体长度可能有所不同，但趋势是一样的。

当然，刚才引用的定理中一定在某处隐含着对这一现象的解释，我们可以尝试利用手头所有的现代方法来提取这一解释。也许我们可以计算出人类控制器所能达到的某些最小 H_∞ 范数，甚至某些最小结构奇异值。当然，在计算中，我们需要充分考虑这类控制器固有的局限，例如反应时间、神经肌肉滞后、肢体惯性，以及许多其他不确定性；这些不确定性涵盖了这样一个事实：人类以及物理世界中几乎所有其他事物都不是有限维、线性且时不变的。

Bode 积分

我将尝试用 Hendrik Bode 可能采用的方式，在频域中解释这些现象。事实证明，这样的解释富有洞见、朴实而清晰，因此比许多现代解释更可取。

首先，注意保持长棍与短棍平衡的区别与不稳定模态的位置有关。从棍的线性化运动方程中很容易看出这一点。在所有质量都集中于棍末端这一简化假设下，这些方程表明存在一个不稳定极点，位于 $\sqrt{g/L}$ 处，其中 g 表示重力加速度，L 表示长度。发散随着 L 减小而变得更快。



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图2。*1986*年*4*月*26*日事故发生后不久的切尔诺贝利核电站。

面对这种不断变化的不稳定性，Bode 积分给出了控制难度的频域量化描述（见边栏）。冒着听起来武断的风险，我认为每一位控制理论研究者和每一位控制工程师都应当知道这些积分，并理解其含义。遗憾的是，我们并非总能把它们教好。

我自身的经历印证了上述最后一点观察。你可以结合自己的经历自行判断。在20世纪1960年代本科和研究生阶段的整个控制理论学习过程中，我只接触过第一个积分，而且仅仅

once, and only in an optional reading of an unassigned chapter in one of the classical textbooks. This integral surfaced for me for the second time in the mid 1970s, referenced in a paper by Isaac Horowitz titled "On the Superiority of Transfer Functions over State-Variable Methods. . . ." It appeared as a perspectives paper in *IEEE Transactions on Automatic Control* amid a certain amount of controversy [2].

The second integral did not surface for me until 1983, in a talk by Jim Freudenberg at an IEEE Conference on Decision and Control in San Antonio [3]. If memory serves, someone pointed out at the time that this result was "just a version of Jensen's theorem," well known in mathematics for a long time. Perhaps this historical reference reduced the value of the result in the minds of some listeners, but it should not have, because the integral explains so much about the difficulties of controlling unstable systems.

A Bode Integral Interpretation

I like to think of Bode's integrals as conservation laws. They state precisely that a certain quantity—the integrated value of the log of the magnitude of the sensitivity function—is conserved under the action of feedback. The total amount of this quantity is always the same. It is equal to zero for stable plant/compensator pairs, and it is equal to some fixed positive amount for unstable ones.

Since we are talking about the log of sensitivity magnitude, it follows that negative values are good (i.e., sensitivities less than unity, better than open loop) and positive values are bad (i.e., sensitivities greater than unity, worse than open loop). So for open-loop stable systems, the average sensitivity improvement a feedback loop achieves over frequency is exactly offset by its average sensitivity deterioration. For open-loop unstable systems, things are worse because the average deterioration is always larger than the improvement. This applies to every controller, no matter how it was designed. Sensitivity improvements in one frequency range must be paid for with sensitivity deteriorations in another frequency range, and the price is higher if the plant is open-loop unstable.

It is curious, somehow, that our field has not adopted a name for this quantity being conserved (i.e., the integrated log of sensitivity magnitude), to put it on a par with some of the great quantities of physics such as mass, momentum, or energy. But since it has not, we are free to choose a name right now. Let me propose that we simply call it *dirt*. It is stuff we would rather not have around; the less we have, the better. I want to choose this name because it lets me liken the job of a serious control designer to that of a ditch digger, as illustrated in Figure 3. He moves dirt from one place to another, using appropriate tools, but he never gets rid of any of it. For every ditch dug somewhere, a mound is deposited somewhere else. This fact is most evident to the ditch digger, because he is right there to see it happen.

In the same spirit, I can also illustrate the job of a more academic control designer with more abstract tools such as linear quadratic Gaussian (LQG), H_∞, convex optimization, and the like, at his disposal. This designer guides a powerful ditch-digging machine by remote control from the safety of his workstation (Figure 4). He sets parameters (weights) at his station to adjust the contours of the machine's digging blades to get just the right shape for the sensitivity function. He then lets the machine dig down as far as it can, and he saves the resulting compensator. Next, he fires up his automatic code generator to write the implementation code for the compensator, ready to run on his target microprocessor.

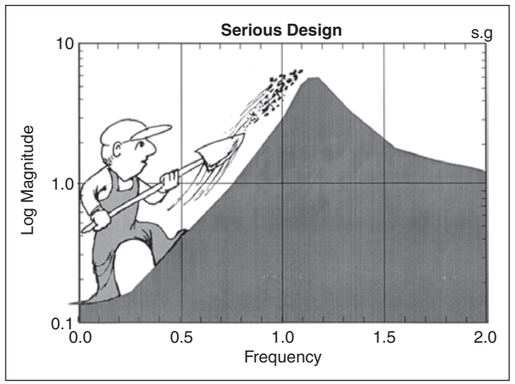


Figure 3. *Sensitivity reduction at low frequency unavoidably leads to sensitivity increase at higher frequencies.*

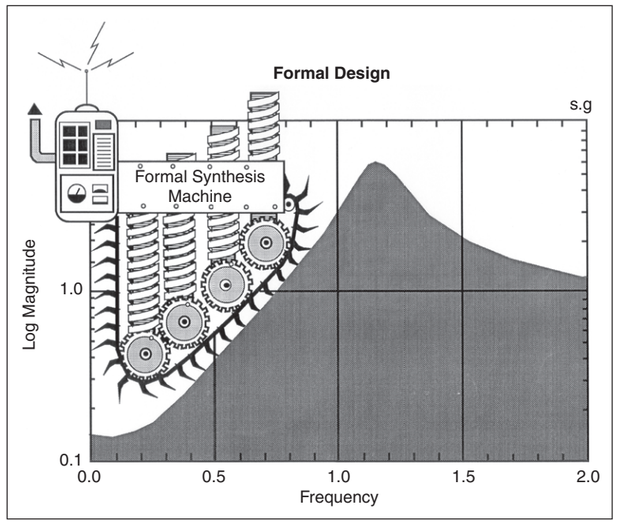


Figure 4. *Sensitivity shaping automated by modern control tools.*

一次，而且仅见于某本经典教科书中一个未被指定阅读的章节的选读内容。我第二次遇到这个积分是在 1970 年代中期，Isaac Horowitz 在一篇题为《论传递函数相对于状态变量方法的优越性……》的论文中引用了它。这篇文章作为一篇观点论文发表于 *IEEE* 自动控制汇刊，并引起了一定争议 [2]。

直到1983年，在圣安东尼奥举行的 IEEE 决策与控制会议上听了 Jim Freudenberg 的一场报告后，我才了解到第二个积分 [3]。如果没记错，当时有人指出，这一结果"只是詹森定理的一个版本"，而该定理在数学界早已广为人知。也许这一历史渊源让一些听众低估了这一结果的价值，但这不应该，因为这个积分解释了控制不稳定系统时遇到的诸多困难。

波德积分的一种解释

我喜欢将波德积分看作守恒定律。它们精确地指出，某个量——灵敏度函数幅值的对数的积分值——在反馈作用下保持守恒。这个量的总量始终不变。对于稳定的被控对象/补偿器组合，它等于零；对于不稳定的组合，它等于某个固定的正值。

由于我们讨论的是灵敏度幅值的对数，因此负值是有利的（即灵敏度小于1，优于开环），而正值是不利的（即灵敏度大于1，劣于开环）。所以，对于开环稳定的系统，反馈回路在整个频率范围内实现的平均灵敏度改善，恰好被其平均灵敏度恶化所抵消。对于开环不稳定的系统，情况更糟，因为平均恶化总是大于平均改善。这适用于每一个控制器，无论它是如何设计的。在一个频率范围内改善灵敏度，必须以另一个频率范围内的灵敏度恶化为代价；如果被控对象开环不稳定，这一代价就更高。

不知为何，我们这个领域竟然还没有为这个守恒量（即灵敏度幅值的对数积分）取一个名字，使其与质量、动量或能量等物理学中的重要物理量并列，这颇令人感到奇怪。不过，既然还没有名字，我们现在就可以自由地取一个。我提议，干脆将它称为泥土。这是我们不希望周围存在的东西；越少越好。我选择这个名字，是因为它让我可以将一位严谨的控制设计者的工作比作挖沟工的工作，如图 3 所示。他使用适当的工具，将泥土从一个地方搬到另一个地方，但从来没有消除任何泥土。每当在某处挖出一条沟，就会在另一处堆起一个土堆。这一事实对挖沟工来说最为显而易见，因为他就在现场，亲眼看着这一切发生。

本着同样的思路，我也可以说明一位更偏学术的控制设计人员的工作，他可以使用线性二次高斯（LQG）、H_∞、凸优化等更抽象的工具。这位设计人员在安全的工作站上，通过遥控操纵一台强大的挖沟机（图 4）。他在工作站上设置参数（权重），调整机器挖掘刀片的轮廓，使灵敏度函数获得恰到好处的形状。然后，他让机器尽可能向下挖掘，并保存由此得到的补偿器。接下来，他启动自动代码生成器，为补偿器编写实现代码，使其准备好在目标微处理器上运行。

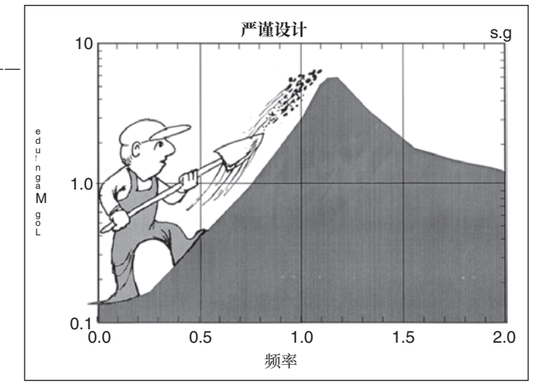


图 3。低频处灵敏度的降低不可避免地会导致较高频率处灵敏度的增加。

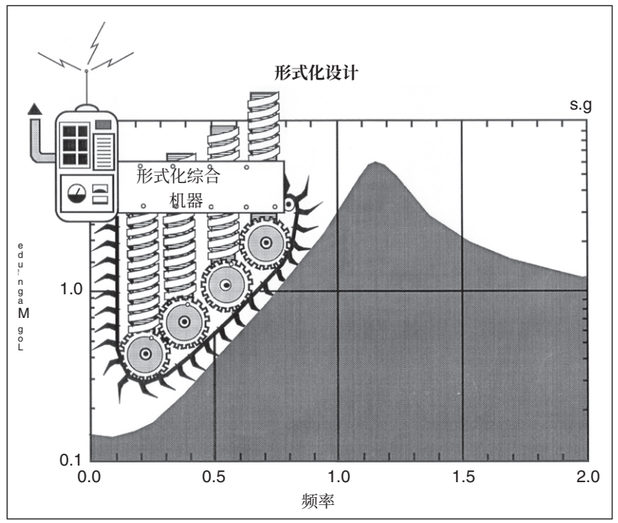


图 4. 由现代控制工具自动实现的灵敏度整形。

He downloads that code automatically to the microprocessor and hits the power-on button on the control system. Indeed, this entire process can become so automatic and insulated that the designer may never look at what has actually happened to his control loop, and all too often the power-on creates a rude surprise.

Available Bandwidth

Many rude surprises that occur in automated design scenarios have to do with excessive bandwidth. The designer unwittingly allows the machine to dig too deeply, piling up dirt at high frequencies where it cannot be supported.

The notion that dirt piled at high frequencies needs support is not taken seriously enough in the theoretical community, even today. For example, an argument is sometimes made that the Bode integrals are not really restrictive because we only seek to dig holes over finite frequency bands. We then have an infinite frequency range left over into which to dump the dirt, so we can make the layer arbitrarily thin. The weakness of this argument is evident from standard classical theory. A thin layer, say with $\ln|s| = \varepsilon$, requires a loop transfer function whose Nyquist diagram falls on a near-unit circle, centered at $(-1 + j0)$ with radius $\approx (1 - \varepsilon)$, over a wide frequency range. This means that the loop cannot simply attenuate at high frequencies but must attenuate in a very precise way. The loop must maintain very good frequency response fidelity over wide frequency ranges.

But a key fact about physical systems is that they do not exhibit good frequency response fidelity beyond a certain bandwidth. This is due to uncertain or unmodeled dynamics in the plant, to digital control implementations, to power limits, to nonlinearities, and to many other factors. Let us call that bandwidth the "available bandwidth," Ω_a, to distinguish it from other bandwidths such as crossover or 3-dB magnitude loss. The available bandwidth is the frequency up to which we can keep $gk(j\omega)$ close to a nominal design and beyond which we can only guarantee that the actual loop magnitude will attenuate rapidly enough (e.g., $|gk| < \delta/\omega^2$). In today's popular robust control jargon, the available bandwidth is the frequency range over which the unstructured multiplicative perturbations are substantially less than unity.

Note that the available bandwidth is not a function of the compensator or of the control design process. Rather, it is an a priori constraint imposed by the physical hardware we use in the control loop. Most importantly, the available bandwidth is always finite.

Given all this, Bode's integrals really reduce to finite integrals over the range $0 \le \omega \le \Omega_a$, i.e,

$$\int_0^{\Omega_a} \ln|s(j\omega)|\, d\omega = \delta \quad \text{stable loops}$$

$$\int_0^{\Omega_a} \ln|s(j\omega)|\, d\omega = \pi \sum_{p \in P} \mathrm{Re}(p) + \delta \quad \text{unstable loops.}$$

All the action of the feedback design, the sensitivity improvements as well as the sensitivity deteriorations, must occur within $0 \le \omega \le \Omega_a$. Only a small error ($\delta$) occurs outside that range, associated with the tail of the complete integrals. Since the details of this tail are not guaranteed by the design, the error can be either positive or negative. The design only guarantees that it will be small.

As an aside, you probably know that other constraints imposed by right-half-plane zeros also give rise to finite integrals, although in different variables [4]. I did not choose to use these additional integrals here because they still require infinite available bandwidth, and I believe that finiteness of available bandwidth is a more significant concept in control than nonminimum phaseness, even though it is not nearly so elegant.

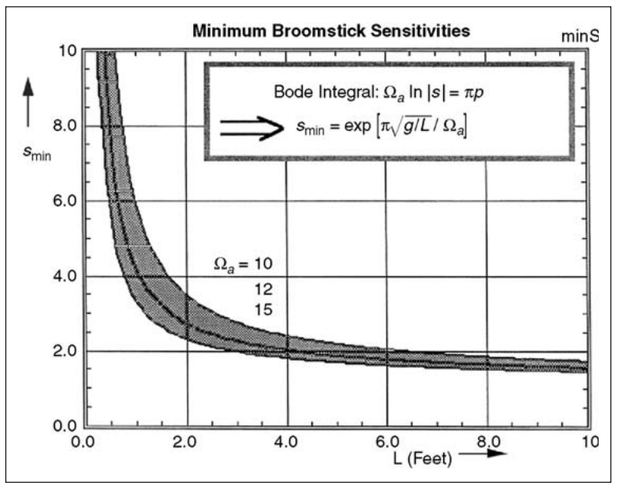


Figure 5. *Sensitivity constraints as a function of broomstick length.*

An Explanation of the Broomstick

These last integrals bring us to the point where we can explain the broomstick balancing illustration in a Bode-like way. First, what is the available bandwidth of the feedback loop in that experiment? Looking at the plant alone, it is fairly high. The stick is stiff, air drag is negligible, and little else prevents the stick from moving as required. The compensator, however, is another matter. Its physical implementation by a human operator has many complex limitations associated with perception, computation, and actuation of limbs. Many years of study and experimentation have gone into the characterization of these limitations, especially for piloting tasks in military airplanes. Since we obviously cannot go through all that here, let us simply agree that

他将该代码自动下载到微处理器，并按下控制系统的通电按钮。事实上，这整个过程可能变得如此自动化、如此与实际隔绝，以至于设计人员可能从未查看自己的控制回路究竟发生了什么，而通电后出现令人措手不及的意外，实在太过常见。

可用带宽

自动化设计场景中出现的许多令人措手不及的意外都与带宽过大有关。设计者不知不觉地让机器挖得过深，将泥土堆积在无法得到支撑的高频处。

即使在今天，理论界对堆积在高频处的泥土需要支撑这一观念仍未给予足够重视。例如，有时有人认为，Bode 积分实际上并不构成限制，因为我们只寻求在有限的频带内挖坑。这样，我们还剩下无限的频率范围可以倾倒泥土，因此可以把土层铺得任意薄。从标准经典理论来看，这一论点的缺陷显而易见。薄土层，例如满足 $\ln|s| = \varepsilon$ 的土层，要求环路传递函数的 Nyquist 图在很宽的频率范围内落在一个接近单位圆的圆上，其圆心为 $(-1 + 0j)$，半径为 $\approx (-1 \varepsilon)$。这意味着环路不能只是在高频处衰减，而必须以非常精确的方式衰减。环路必须在很宽的频率范围内保持很高的频率响应保真度。

但物理系统的一个关键事实是，超过某一带宽后，其频率响应便无法保持良好的保真度。这是由被控对象中不确定或未建模的动力学、数字控制实现、功率限制、非线性以及许多其他因素造成的。我们将该带宽称为“可用带宽”Ω_a，以区别于交越带宽或3-dB幅值衰减带宽等其他带宽。可用带宽是这样一个频率上限，在此频率以内

我们可以使$gk\,j(\,)\omega$ 保持接近标称设计，而超过该频率后，我们只能保证实际环路幅值衰减得足够快（例如，$|gk| < \delta/\omega^2$）。用当今流行的鲁棒控制术语来说，可用带宽是非结构化乘性扰动显著小于1的频率范围。

请注意，可用带宽并非补偿器或控制设计过程的函数。相反，它是我们在控制回路中使用的物理硬件所施加的先验约束。最重要的是，可用带宽始终是有限的。

考虑到以上所有因素，Bode 积分实际上可化为在 range$0 \le \omega \le \Omega_a$ 上的有限积分，即，

$$\int_0^{\Omega_a} \ln|s(j\omega)|\, d\omega = \delta \quad \text{stable loops}$$

$$\int_0^{\Omega_a} \ln|s(j\omega)|\, d\omega = \pi \sum_{p \in P} \mathrm{Re}(p) + \delta \quad \text{unstable loops.}$$

反馈设计的所有作用，包括灵敏度的改善和恶化，都必须发生在 $0 \le \omega \le \Omega_a$ 内。在该范围之外，仅会出现一个小误差（）δ，它与完整积分的尾部有关。由于设计无法保证该尾部的具体情况，误差可能为正，也可能为负。设计仅能保证误差很小。

顺便说一句，你可能知道，右半平面零点施加的其他约束也会产生有限积分，不过所用的变量不同 [4]。我没有选择在这里使用这些附加积分，因为它们仍然要求无限的可用带宽，而且我认为，在控制中，可用带宽的有限性比非最小相位性更为重要，尽管这一概念远不如后者优雅。

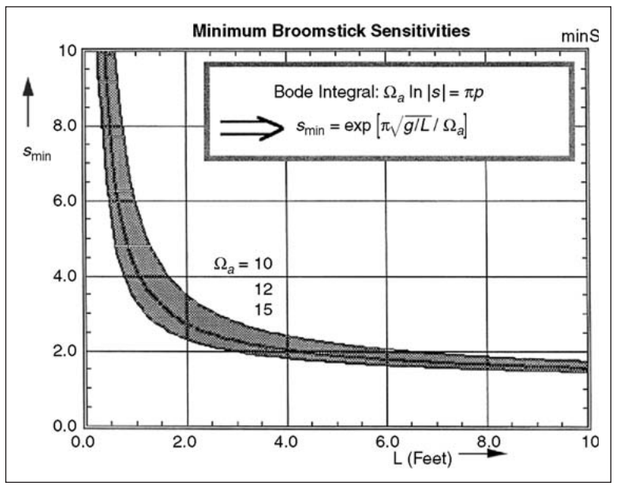


图 5. 灵敏度约束随扫帚杆长度的变化。

扫帚杆平衡的解释

上述最后几个积分使我们能够以类似伯德分析的方式解释扫帚杆平衡的示例。首先，该实验中反馈回路的可用带宽是多少？仅看被控对象，其带宽相当高。杆是刚性的，空气阻力可以忽略，而且几乎没有其他因素妨碍杆按要求运动。然而，补偿器则是另一回事。由操作者实现的补偿器在感知、计算和肢体动作方面存在许多复杂的限制。多年来，人们开展了大量研究和实验来表征这些限制，尤其是针对军用飞机的驾驶任务。显然，我们无法在此逐一讨论这些内容，因此，让我们姑且认同

the compensator is good for a frequency range up to ≈2 Hz (say, 10-15 rad/s) and that its control strategy is to keep sensitivity as small as possible over that range (i.e., keep the loop's Nyquist diagram as far away from the critical point as possible everywhere).

The curves in Figure 5 show how well the closed-loop system will perform. These curves are simply restatements of the finite Bode integral, with sensitivity assumed equal to a constant minimum achievable value over $0 \le \omega \le \Omega_a$ and equal to unity elsewhere. The minimum achievable value is an exponential function of the ratio of unstable pole to available bandwidth (p/Ω_a) and with p given by $\sqrt{g/L}$, it is an explicit function of the broomstick length. Notice that a dramatic increase in sensitivity occurs below a foot and a half. These large sensitivities put the loop close to the critical point, and even minor imperfections in the implementation will cause instability. That, quite simply, is the reason we humans have trouble balancing short sticks, and other controllers with similar available bandwidth limits have trouble as well. This reason follows directly from the most fundamental conservation laws of feedback, Bode's integrals, and needs none of our modern mathematical machinery. More importantly, none of the modern results overcome these limitations. They can (and do!) hide them from us, but they do not remove them.

The X-29 Airplane Story

Although the broomstick-balancing illustration itself is only a toy, it represents some very serious physical systems with similar dynamics. In addition to the launch booster analogy already mentioned, there is also a direct analogy to unstable airplanes. As an example of the latter, I would like to tell a story about the X-29, the forward-swept-wing research airplane shown in Figure 6.

This airplane was built to demonstrate basic aerodynamic performance improvements that might be gained from new composite materials tailored specifically for aerodynamic efficiency. The wings are swept forward instead of aft, so that bending enhances lift instead of decreasing it at high angles of attack; a large canard surface is placed close in to take advantage of favorable interactions with the wing; and there are several other features as well. Once demonstrated, these technologies will make their way into next-generation airplanes.

The airplane was built by Grumman Aircraft Company, under U.S. Air Force, DARPA, and NASA sponsorship, and underwent successful flight tests for several years. Honeywell supplied the control hardware and software. Control laws were designed at Grumman, with supporting design activity at Honeywell, as well as several other places.

Although the airplane's controls were not a major focus of the flight demonstration, they are of interest here because they illustrate how extreme our worship of formal methods has become.You see, all of the various control design teams used modern digging machines early in the design process. As a result, we were well insulated from the fundamental difficulties imposed by the airplane's violent open-loop instability. We discovered only at the last moment that the vehicle was almost too unstable to control with the given hardware.

We will consider unstable to be synonymous with dangerous.

I want to relate this story by talking first about what makes the airplane unstable, then about the hardware features that restrict available bandwidth, and finally, I will put these together with Bode's integral to show the airplane's fundamental control limitations.

Static Instability

An airplane is open-loop unstable when its center of pressure (cp, the effective point of action of lift forces) is located ahead of its center of gravity (cg). Since lift forces grow in direct proportion to pitch (nose up or down) attitude, any initial pitching motion changes lift, which acts through the cp–cg offset to produce moments in the same direction, and the attitude diverges. In aeronautical circles, this condition is called static instability. The associated linearized dynamics look very much like broomstick-balancing equations. There are two roots—one stable, one unstable—approximately equal in magnitude.

Static instability does not happen arbitrarily. It is deliberately designed into an airplane by locating lifting surfaces



Figure 6. *NASA X-29 forward-swept-wing aircraft (photo courtesy of NASA).*

该补偿器适用的频率范围上限为 ≈2 Hz（例如，10-15 rad/s），其控制策略是在该范围内使灵敏度尽可能小（即，使环路的奈奎斯特图在所有位置都尽可能远离临界点）。

图 5 中的曲线表明闭合 -环系统的性能如何。这些曲线只是对有限频段波德积分的重新表述，其中假设灵敏度等于可达到的恒定最小值 over$0 \le \omega \le \Omega_a$，在其他频段则等于一。可达到的最小值是一个指数函数，其自变量为不稳定极点与可用带宽之比（$/\Omega p_a$），并且由于 p 由下式给出：$gL/$，因此它是扫帚柄长度的显式函数。请注意，当长度低于一英尺半时，灵敏度会急剧增大。这些高灵敏度使环路接近临界点，即使实现过程中存在微小缺陷，也会导致不稳定。很简单，这就是我们人类难以使短棍保持平衡的原因，具有类似可用带宽限制的其他控制器也同样面临困难。这一原因直接源于反馈最基本的守恒定律——波德积分，无须借助任何现代数学工具。更重要的是，任何现代成果都无法克服这些限制。它们能够（而且确实会！）使我们看不到这些限制，但无法消除这些限制。

尽管这架飞机的控制系统并非飞行演示的主要重点，但这里仍值得关注，因为它们说明了我们对形式化方法的崇拜已经到了何等极端的地步。要知道，各个控制设计团队都在设计过程的早期使用了现代挖掘机械。因此，我们被很好地隔绝于

我们将把不稳定视为危险的同义词。

飞机剧烈的开环不稳定性所造成的根本性困难。直到最后一刻，我们才发现，这架飞机的不稳定程度几乎超出了给定硬件所能控制的范围。

我想先谈谈是什么使飞机不稳定，再谈谈限制可用带宽的硬件特性，最后将这些因素与 Bode 积分结合起来，说明飞机控制的根本限制，以此讲述这个故事。

静不稳定性

当飞机的压力中心（cp，即升力的有效作用点）位于其重心（cg）前方时，飞机在开环状态下是不稳定的。由于升力随俯仰（机头向上或向下）姿态成正比变化，任何初始俯仰运动都会改变升力，而升力通过 cp–cg 偏距产生同方向的力矩，使姿态发散。在航空领域，这种情况称为静不稳定性。相应的线性化动力学方程与扫帚杆平衡方程非常相似。有两个根——一个稳定，一个不稳定——其大小近似相等。

静不稳定性并非任意发生。通过布置升力面，飞机的静不稳定性被有意设计出来

X-29 飞机的故事

虽然平衡扫帚柄的演示本身只是一个玩具，但它代表了一些具有相似动力学特性的、非常严肃的物理系统。除了前面提到的运载火箭助推器类比之外，它还可以直接类比不稳定飞机。作为后者的一个例子，我想讲一讲 X-29 的故事，它是图 6 所示的前掠翼研究飞机。

建造这架飞机是为了验证，利用专门针对气动效率设计的新型复合材料，可能获得的基本气动性能提升。机翼采用前掠而非后掠，使得在大迎角下，弯曲会增加升力而不是降低升力；较大的鸭翼布置在靠近机翼的位置，以利用其与机翼之间有利的相互作用；此外，它还具有其他一些特点。一旦得到验证，这些技术将应用于下一代飞机。

这架飞机由 Grumman Aircraft Company 在美国空军、DARPA 和 NASA 的资助下制造，并成功进行了数年的飞行试验。Honeywell 提供了控制硬件和软件。控制律由 Grumman 设计，Honeywell 及其他几家机构也开展了配套设计工作。



图 6. *NASA X-29* 前掠翼飞机（照片由 *NASA* 提供）。

and distributing mass appropriately. In the years before full-authority automatic flight controls, most airplane designs constrained the placement of these elements to ensure stability over all flight regimes and all loading conditions. (The Wright brothers' airplane is a notable exception.) Today, however, when automatic controls are accepted and trusted in so many applications, there are important benefits to be gained by making the basic design unstable. For instance, note that a stable airplane requires a tail to balance the moment produced by lift acting through the cp – cg offset. The force produced by the tail is down, opposing the lift and making the overall configuration aerodynamically less efficient. The airplane also needs to carry the weight of the tail, which reduces payload. The less stable an airplane is, the smaller these performance and weight penalties. There are also other reasons for wanting instability, having to do with maneuverability and speeds of command response.

In the case of the X-29, the benefits of instability were desired in the transonic and supersonic flight regimes, so the airplane was designed to be modestly unstable in those regimes. Unfortunately, there is a basic aerodynamic phenomenon that moves the center of pressure of a lifting surface dramatically aft as speeds go from sub- to supersonic. As a result, the X-29's slight instability at supersonic speeds turns into a much more dramatic instability at subsonic speeds. Indeed, the airplane's real pole is as large as +6 rad/s in some flight regimes.

Recalling the broomstick's formula ($p = \sqrt{g/L}$), flying this airplane manually corresponds to balancing a 1-ft-long stick. It is not something we can expect a pilot to do without automatic assistance, at least for very long.

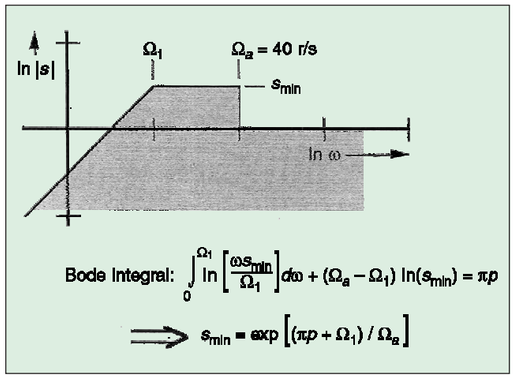


Figure 7. *Prototype X-29 sensitivity function.*

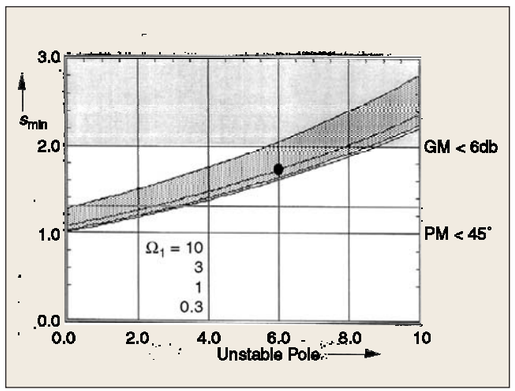


Figure 8. *Minimum X-29 sensitivities.*

X-29 Available Bandwidth

As with the broomstick, control difficulties associated with the X-29's unstable pole should not arise merely because the pole is large. Rather, they should arise if the pole is large compared with available bandwidth. Some of the major hardware elements in the control loop that limit this bandwidth include the following:

- *Sensors*: rate gyros and accelerometers, used for inner-loop stabilization. Their bandwidths, measured in the usual 3-dB-gain sense, are typically 120 rad/s or more.
- *Control processors*: digital computer systems sampling sensor data and computing surface commands at 80 Hz. This update rate can pass signals with good fidelity up to 30-40 rad/s (two to three samples per radian).
- *Actuators*: high-pressure hydraulic systems with servos to position each aerodynamic control surface. For the pitch axis, control surfaces include the canard, inboard and outboard flaperons along the wing, and strakes at the tail. The servo bandwidths, again measured by 3-dB gain, are approximately 20 rad/s. Basically, each of the servos is a position feedback loop around a hydraulic ram (an integrator). This gives a dominant first-order-lag response that retains its characteristics up to perhaps 70-80 rad/s, where valve dynamics, fluid compressibility, and local structural dynamics begin to take effect. The bottom line is that the available bandwidth of actuators is approximately 70 rad/s, at least for small signals.
- *Aerodynamics*: flow conditions around the airplane that map its geometric configuration (e.g., net orientation and control surface positions) into forces and moments. At low frequencies, this map is algebraic and well known, but because air itself has mass and momentum, the map becomes dynamic and poorly known on time scales comparable to characteristic length divided by velocity. For a 2-ft surface traveling at 200 ft/s, that time scale is 0.01 s, so the available bandwidth for aerodynamics is approximately 100 rad/s.
- *Airframe:* the mechanical structure linking the attachment points of actuators to the attachment points of sensors. (These are the important structural points affecting control. Other disciplines are, of course, interested in additional attachment points, such as where the wings are attached, where the pilot sits, etc.) The structure is rigid at low frequencies, but it begins to bend and flex dramatically at higher fre-

并适当分配质量。在全权限自动飞行控制出现之前的年代，大多数飞机设计都对这些要素的布置施加约束，以确保在所有飞行状态和所有载荷条件下都具有稳定性。（莱特兄弟的飞机是一个显著的例外。）然而，如今自动控制已在众多应用中得到接受和信赖，使基本设计具有不稳定性能够带来重要的益处。例如，请注意，稳定的飞机需要尾翼来平衡升力通过 cp–cg 偏距作用而产生的力矩。尾翼产生的力向下，与升力相反，使整体构型的气动效率降低。飞机还需要承载尾翼的重量，从而减少有效载荷。飞机的稳定性越低，这些性能和重量方面的代价就越小。希望飞机具有不稳定性还有其他原因，这些原因与机动性和指令响应速度有关。

对于 X-29，设计者希望在跨声速和超声速飞行状态下获得不稳定性带来的益处，因此将飞机设计为在这些状态下具有适度的不稳定性。遗憾的是，存在一种基本的空气动力学现象，会使升力面的压力中心

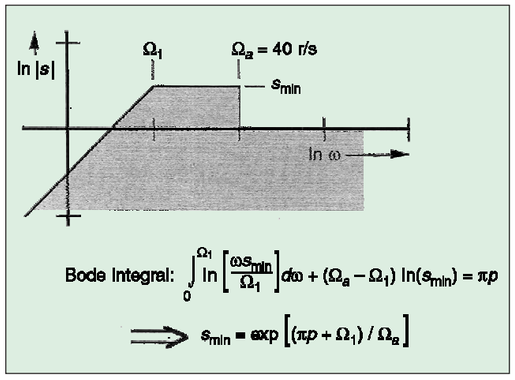


图 7. *X-29* 原型机灵敏度函数。

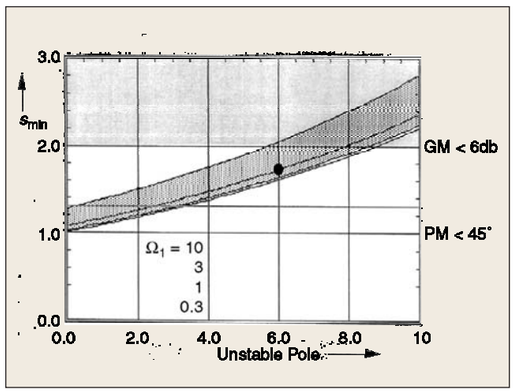


图 8. *X-29* 最小灵敏度。

随着速度从亚声速增至超声速而大幅后移。因此，X-29 在超声速下的轻微不稳定性，在亚声速下变成了严重得多的不稳定性。事实上，在某些飞行状态下，该飞机的实极点高达 +6 rad/s。

回顾扫帚杆的公式(/)*p g L*= ., 手动驾驶这架飞机相当于使一根 1-ft 长的杆保持平衡。我们不能指望飞行员在没有自动辅助的情况下做到这一点，至少不能长时间做到。

X-29 可用带宽

与扫帚杆一样，与 X-29 的不稳定极点相关的控制困难，不应仅仅因为极点很大而出现。相反，当极点相对于可用带宽而言很大时，才应出现这些困难。控制回路中限制该带宽的一些主要硬件包括以下几项：

• 传感器：速率陀螺仪和加速度计，用于内环增稳。按通常的 3-dB 增益标准衡量，其带宽通常为 120 rad/s 或更高。• 控制处理器：以 80 Hz 对传感器数据进行采样并计算舵面指令的数字计算机系统。此更新速率可高保真地传递频率高达 30-40 rad/s 的信号（每弧度采样两到三次）。

• 执行机构：采用伺服机构定位各气动操纵面的高压液压系统。对于俯仰轴，操纵面包括鸭翼、沿机翼分布的内侧和外侧襟副翼，以及尾部边条翼。同样以 3-dB 增益衡量，伺服机构的带宽约为 20 rad/s。基本上，每个伺服机构都是围绕液压作动筒（一个积分器）构成的位置反馈回路。这产生了占主导地位的一阶滞后响应，其特性可保持至大约 70-80 rad/s，此时阀门动态、流体可压缩性和局部结构动态开始起作用。因此，执行机构的可用带宽约为 70 rad/s，至少对于小信号而言是如此。

• 空气动力学：飞机周围的流动条件，将其几何构型（例如整体姿态和操纵面位置）映射为力和力矩。在低频下，这种映射是代数关系，且已得到充分认识，但由于空气本身具有质量和动量，当时间尺度与特征长度除以速度所得的值相当时，这种映射就会变为动态关系，且认识尚不充分。对于以 200 ft/s 运动的 2-ft 翼面，该时间尺度为 0.01 s，因此空气动力学的可用带宽约为 100 rad/s。

• 机体结构：连接执行器安装点与传感器安装点的机械结构。（这些是影响控制的重要结构点。当然，其他学科还关注其他安装点，例如机翼的连接位置、飞行员的座位位置等。）该结构在低频下呈刚性，但在较高频率下开始发生显著的弯曲和挠曲

quencies. Typical fighter-sized airplanes have a first fuselage bending mode at ≈ 7.0 Hz (40 rad/s). The frequency of this mode, as well as its mode shape and damping characteristics, move around substantially with mass distribution (fuel, payloads) and with dynamic loading (maneuvers). This makes it very difficult to maintain good frequency response fidelity beyond the first mode frequency.

This list shows that the most severe limitation on available bandwidth for the X-29 is approximately 40 rad/s and comes from the airplane's mechanical structure and from the sampling rate of its computers. I have elected to describe the various other items, at least briefly, to emphasize the point that real physical systems have a multitude of limitations on available bandwidth, not just one or two that can be easily pushed out or removed. We cannot simply ignore them, as we often do in formal theories. Instead, we should fashion theories to clearly expose the consequences of these limitations, and we should honestly tell ourselves and our employers what the consequences are.

As mentioned earlier, in the initial X-29 designs, the various design teams used some of the formal digging machines, which hide these consequences. Not until late in the design process, and not without heated arguments among designers, did we all agree that they are indeed real and unavoidable.

X-29 Limitations

Using Bode's integral, the limitations are easy to demonstrate. We have a plant with an unstable pole, $p = 6$ rad/s, and with an available bandwidth, $\Omega_a = 40$ rad/s. We want to achieve a sensitivity function shaped approximately like the prototype shown in Figure 7. This prototype has small sensitivity at low frequencies, rising with $a+1$ slope up to Ω_1. It then stays flat up to Ω_a, so as to pay as small a penalty as possible, consistent with Bode's integral. Finally, it returns to unity (0 dB) beyond Ω_a. A formula for the smallest sensitivity penalty, $s_{\min}$, is easy to derive directly from the integral and is also shown in the figure.

Some $s_{\min}$ curves based on this formula are shown in Figure 8. We see that the penalty level rises as the airplane becomes more unstable and also as the frequency Ω_1, up to which we want good performance, increases. The design point for the X-29 is shown at $p = 6$ rad/s and $\Omega_1 = 3$ rad/s. The resulting smallest sensitivity penalty is ≈ 1.75. Classical designers will immediately recognize this to be marginal because traditional phase margins will not be satisfied. These margins are given in terms of $s_{\min}$ by well-known formulas [e.g., $PM = 2\sin^{-1}(1/2s_{\min})$]. Boundaries for standard military flight control specifications, a 45°-phase margin and a 6-dB gain margin, are shown in the figure.

Notice that we have not actually done any design work. We have made no reference to design methods or to big ditch-digging machines. Instead, we have just used a basic Bode integral calculation to determine that the situation is marginal.

To confirm that the situation is marginal, Figure 9 shows a Bode diagram of a loop transfer function corresponding to a realizable version of the prototype shape in Figure 7. This realizable version was found by taking a fourth-order approximation of Figure 7 (basically rounding the corners), selecting one free parameter (the new $s_{\min}$) to satisfy Bode's integral, and then solving explicitly for the loop transfer function (i.e., $gk(s) = s^{-1}(s) - 1$). Note that the resulting loop is well rolled off at the 40-rad/s available bandwidth and that 35° is the largest phase margin achieved by the prototype.

Based on what we have just shown, no controller can make $s_{\min}$ smaller under the given constraints. Thus, we should not be surprised that no acceptable controller was ever found, even with several design teams working on the problem. Indeed, the airplane flies today only because special specification relief was granted. As described in [5], the airplane's stability margins were actually measured explicitly in flight. Two curves from these flight tests are shown as Bode diagrams in Figure 10, one with slightly different control gains than the other. Note the remarkably close correspondence between the Bode-integral-derived prototype and the actual final flight control loop.

Based on this experience with the X-29, it is unlikely that airplanes now being developed, such as the Saab JAS-39, the Advanced Technology Fighter (ATF), and the National Aerospace Plane (NASP X-30), will be so violently unstable. A good rule of thumb, seen from Figure 8, is that available bandwidth should exceed the airplane's unstable pole by at least a factor of ten.

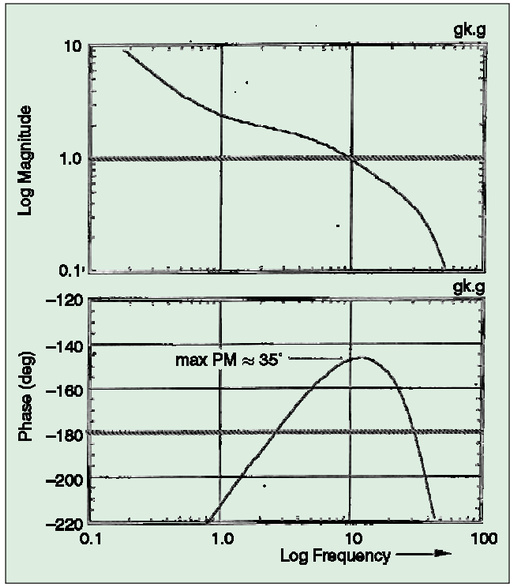


Figure 9. *Prototype Bode diagram for the X-29.*

。典型战斗机尺寸的飞机，其机身一阶弯曲模态的频率为 ≈7.0 Hz (40 rad/s)。该模态的频率及其振型和阻尼特性会随质量分布（燃油、有效载荷）和动态载荷（机动）发生显著变化。这使得在一阶模态频率以上保持良好的频率响应保真度非常困难。

这份清单表明，X-29 的可用带宽受到的最严苛限制约为 40 rad/s，这一限制来自飞机的机械结构及其计算机的采样率。我选择至少简要描述其他各项限制，是为了强调：实际物理系统的可用带宽受到众多限制，而不只是受到一两个可以轻易放宽或消除的限制。我们不能像在形式化理论中经常做的那样，简单地忽略它们。相反，我们应当构建能够清楚揭示这些限制所带来后果的理论，并且应当如实告诉自己和雇主这些后果是什么。

如前所述，在 X-29 的初始设计中，各设计团队使用了一些形式化的挖掘机器，而这些机器掩盖了上述后果。直到设计过程的后期，而且是在设计人员之间经历了激烈争论之后，我们才一致承认，这些后果确实存在且不可避免。

X-29 的限制

利用 Bode 积分，很容易说明这些限制。我们的被控对象具有一个不稳定极点，$p = 6$ rad/s，可用带宽为 $\Omega_a = 40$ rad/s。我们希望得到一个形状近似于图 7 所示原型的灵敏度函数。该原型在低频处的灵敏度较小，以 $a+1$ 的斜率上升，直至 Ω_1。随后保持平坦，直至 Ω_a，以便在满足 Bode 积分的前提下，使代价尽可能小。最后，在 Ω_a 以上恢复为单位值（0 dB）。最小灵敏度代价 $s_{\min}$ 的公式很容易直接由该积分推导得到，图中也给出了这一公式。

图 8 给出了基于该公式的一些 $s_{\min}$ 曲线。可以看到，随着飞机变得更加不稳定，以及我们希望保持良好性能的频率上限 Ω_1，增大，代价水平也随之上升。X-29 的设计点位于 $p = 6$ rad/s 和 $\Omega_1 = 3$rad/s 处。由此得到的最小灵敏度代价为 ≈1.75。经典控制设计人员会立即意识到，这已处于临界状态，因为它无法满足传统的相位裕度要求。这些裕度可通过众所周知的 formulas[e.g. 用 $s_{\min}$ 表示，$PM = 2\sin^{-1}(1/2s_{\min})$]。图中给出了标准军用飞行控制规范的边界，即 45° 相位裕度和 6-dB 增益裕度。

请注意，我们实际上并未进行任何设计工作。我们没有提及设计方法，也没有提及大型挖沟机械。相反，我们只是利用基本的 Bode 积分计算，确定该情况处于临界状态。

为确认该情况处于临界状态，图 9 给出了一个环路传递函数的 Bode 图，该传递函数对应于图 7 中原型形状的一个可实现版本。这个可实现版本是通过对图 7 进行四阶近似（基本上就是将拐角圆滑化），选择一个自由参数（新的$s_{\min}$）以满足 Bode 积分，然后显式求解环路传递函数（即 $gk = s/s(1) - 1$）得到的。请注意，所得环路在 40-rad/s 的可用带宽处具有良好的滚降特性，且 35° 是该原型所能达到的最大相位裕度。

根据我们刚才展示的结果，在给定约束下，没有任何控制器能够使 $s_{\min}$ 更小。因此，即使有多个设计团队致力于解决这一问题，始终未能找到可接受的控制器也不足为奇。事实上，这架飞机如今能够飞行，仅仅是因为获准特别放宽了规范要求。如 [5],所述，飞机的稳定裕度实际上是在飞行中明确测量的。图 10 以波德图的形式展示了这些飞行试验中的两条曲线，其中一条所采用的控制增益与另一条略有不同。请注意，由波德积分推导出的原型与实际最终飞行控制回路之间的一致程度非常高。

根据 X-29 的这一经验，目前正在研制的飞机，如 SaabJAS-39、先进技术战斗机（ATF）和国家空天飞机（NASP X-30），不太可能如此剧烈地不稳定。从图 8 可以看出，一条实用的经验法则是，可用带宽应至少为飞机不稳定极点的十倍。

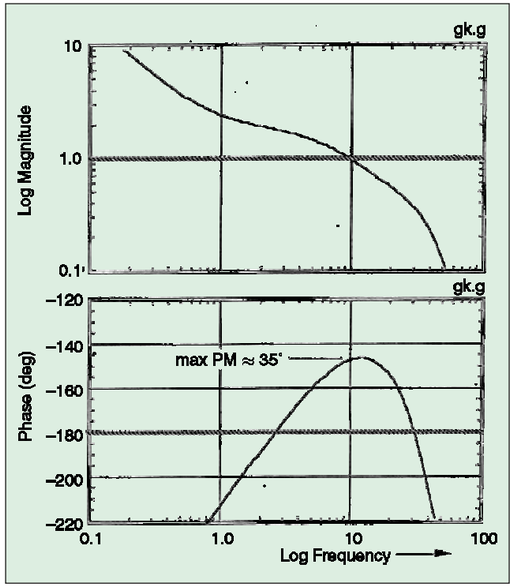


图 9. *X-29* 的原型波德图。

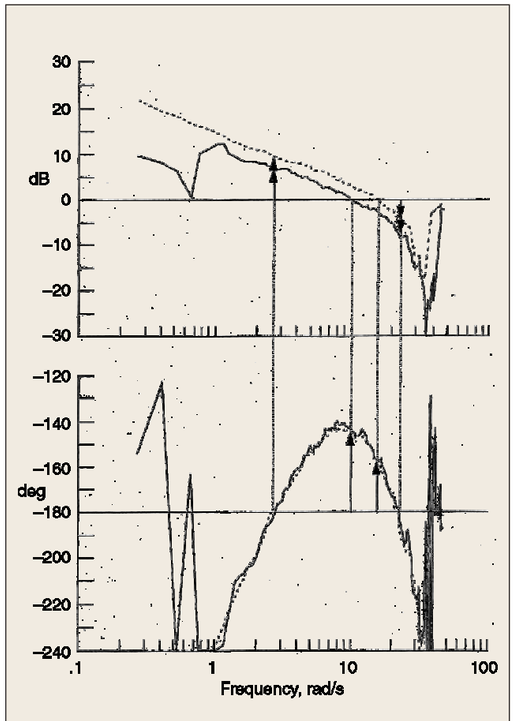


Figure 10. *X-29 flight data (courtesy Mr. J. Gera, NASA).*

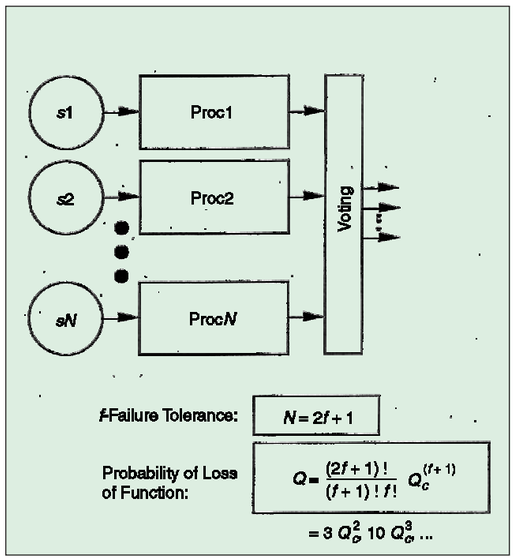


Figure 11. *Basic redundant control architecture.*

Fact 2: Operationally Critical Controllers

We now turn to the second fact about unstable/dangerous systems that I want to emphasize, namely, that controllers for these systems are operationally critical. Controllers must work properly for the systems to be safe. In the case of airplanes, such controllers are called "flight critical." Bluntly stated, this means that if the controller fails, you eject or you die.

Although this may be obvious, it is well worth some discussion. In effect, today's control engineers working with dangerous applications must design and build control systems that are so nearly perfect that we are willing to stake our fortunes and sometimes our lives on them.

More often than not, the backbone of a control system involves electronic equipment, and it is safe to say that most of us have had experiences with such equipment—with our TVs, stereo systems, personal computers, etc.—that do not encourage excessive risk taking. Even the very best equipment fails at a rate of 10^{-3} to 10^{-4} failures per operating hour. The job of control hardware designers, therefore, is to build very reliable systems out of unreliable components. This can be done with careful use of redundant elements arranged in architectures that let the overall hardware/software system maintain its function even though components within it have failed. Indeed, an entire engineering subspecialty has evolved to handle such design problems. It is not my intent to cover much of this subspecialty here, but we should be aware of some of its basic concepts and issues. Certainly we need to recognize that this work is every bit as important to control as are the control theories and design tools that are our own specialty.

Perhaps the most basic architectural concept for building reliable systems out of unreliable components is shown in Figure 11. We simply replicate the entire hardware set of a control loop (sensors, processors, and input/output devices) in several channels and arrange for some sort of voting logic to keep only good channels in control. A simple majority-voting scheme suggests that it takes $2f+1$ channels to still be operational after f failures. This is because there are still $f+1$ channels that agree and can outvote the f failed ones. So if a system must be operational after one failure (so-called fail-op), we get a three-channel or triplex architecture. If operation is required after two arbitrary failures (fail-op squared), we get a five-channel architecture, and so on. (In practice, it is customary to claim fail-op squared capability even with only four channels. This relies on an unstated assumption that the two failures are not simultaneous and that the first has been isolated before the second occurs.)

The statistical reliability of the architecture also increases with the number of channels, as given by the formulae in the figure. Q_c is the failure rate of a channel and Q is the overall failure rate of the implementation.

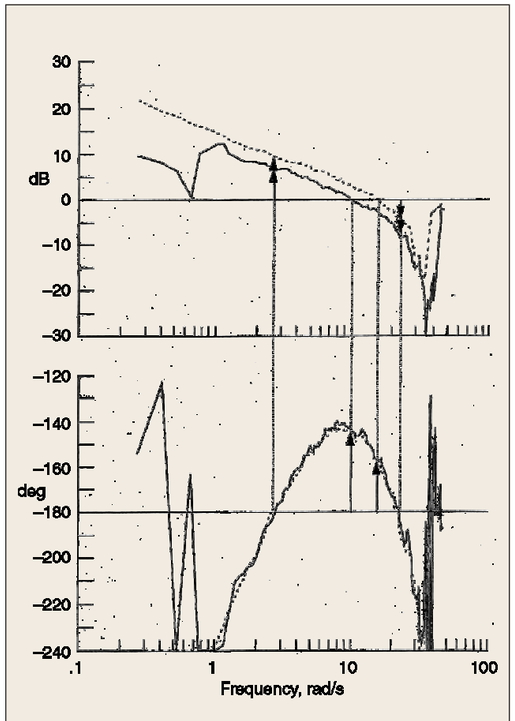


图 10. *X-29* 飞行数据（由 *NASA* 的 *J. Gera* 先生提供）。

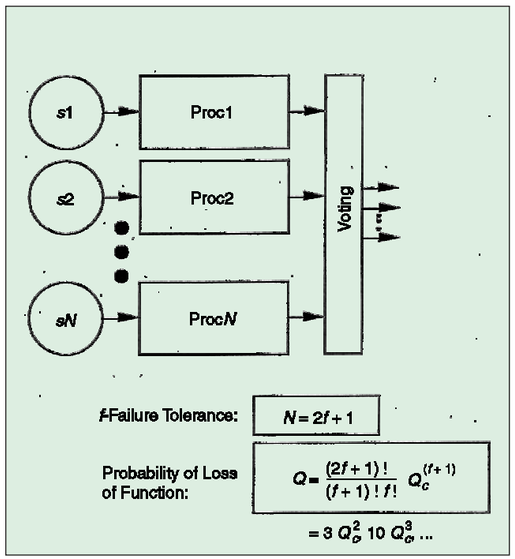


图 11. 基本冗余控制架构。

事实 2：对运行至关重要的控制器

现在转向我想强调的关于不稳定/危险系统的第二个事实，即这些系统的控制器对其运行至关重要。控制器必须正常工作，系统才能安全。对于飞机，这类控制器被称为"飞行关键"控制器。直截了当地说，这意味着如果控制器失效，你要么弹射逃生，要么死亡。

尽管这一点可能显而易见，但仍很值得讨论。实际上，如今从事危险应用的控制工程师必须设计并构建近乎完美的控制系统，完美到我们愿意将自己的财产，有时甚至生命，都托付给它们。

控制系统的核心往往涉及电子设备，而且可以肯定地说，我们大多数人使用这类设备——电视、立体声音响系统、个人计算机等——的经历，并不会鼓励我们过度冒险。即使是最好的设备，每运行一小时也会发生 10^{-3}到 10^{-4}次故障。因此，控制硬件设计人员的工作就是用不可靠的元件构建非常可靠的系统。通过审慎使用冗余元件，并将其组织成适当的架构，即使系统内部的元件发生故障，整个硬件/软件系统仍能维持其功能，从而实现这一目标。事实上，为处理此类设计问题，已经发展出一个完整的工程专业分支。我无意在这里过多介绍这一专业分支，但我们应当了解它的一些基本概念和问题。当然，我们需要认识到，这项工作对于控制的重要性，丝毫不亚于我们自身专长领域中的控制理论和设计工具。

利用不可靠的部件构建可靠系统，最基本的架构概念或许如图 11 所示。我们只需在多个通道中复制控制回路的整套硬件（传感器、处理器和输入/输出设备），并设置某种表决逻辑，使控制仅由正常通道执行。简单的多数表决方案表明，在发生 f 次故障后，仍需有 takes2 $+1f$个通道保持运行。这是因为仍有 f $+1$个通道结果一致，其票数能够超过那 f 个发生故障的通道。因此，如果系统必须在一次故障后仍能运行（即所谓的 fail-op），就需要三通道或三重架构。如果要求系统在任意两次故障后仍能运行（fail-op squared），就需要五通道架构，以此类推。（在实践中，通常即使只有四个通道，也会声称具备 fail-op squared 能力。这依赖于一个未明确说明的假设：两次故障并非同时发生，而且在第二次故障发生之前，第一次故障已被隔离。）

该架构的统计可靠性也随通道数量的增加而提高，如图中的公式所示。Q_c 为单个通道的故障率，Q 为该实现的总体故障率。

Voting

Before this basic concept sinks in too firmly, let me caution that the picture presented in Figure 11 hides most of the difficulties and issues that make redundancy management a challenging engineering discipline. In particular, the figure makes reference to "some sort of voting scheme" to distinguish good channels from bad. The scheme is not shown, but consider how it might be built. The basic idea is to compare outputs of various channels and to proceed with the majority. However, the outputs are not discrete yes-or-no votes, but are digital words, 8 to 12 bits long. In separate channels, the words will be different because the sensors read different signals and processors run at different rates, and perhaps for other reasons as well. So we cannot simply vote, but must make comparisons against preset thresholds instead.



Figure 12. *Airbus A320.*

This raises the first big issue: How large must the thresholds be? Consider a case where each processor implements a control law that includes a sophisticated control algorithm—say, an adaptive law with explicit identification (e.g., parameter estimates evolving according to $d\phi/dt = e^T\phi + \cdots$). Each channel executes this law with different noisy sensor data. The channel differences will then consist of free integrations driven by noise, and from our first course in stochastic processes, these differences behave like Brownian motion and will exceed any particular threshold infinitely often. This is not good for voting!

Note that the fancy control law is not the culprit here. The same problem arises for any algorithm with free integrations or unstable dynamics, as well as for control laws with mode logic, saturation protections, and other discrete switches. It arises with any control algorithm for which small input differences can produce large output differences.

A common solution to this voting issue is to utilize cross-channel communication to "equalize" all channels (e.g., force them to synchronize clocks and to process identical data). Voting then reduces to simple bit-by-bit comparisons. Unfortunately, this solution only peels back the first layer of difficulty, because the communications hardware needed to equalize channels must also be very reliable. This requires still more levels of redundancy, fortunately restricted to fewer components [6]. Even with appropriately reliable communications, however, there remain certain failures that cannot be detected, namely those that produce identical symptoms in each channel.

Failures with identical symptoms are called "generic faults." At first glance they would appear to be unlikely (two or more identical failures at the same time), until we realize that undiscovered design errors in hardware or software have precisely this characteristic. In fact, the undeniable possibility of such errors has caused most bit-by-bit synchronous architectures to also carry one or more backup channels, using different control laws, different hardware, and different software. The X-29, for example, has a triplex digital system as its primary controller, but carries three backup channels with very basic control laws implemented in analog electronics. The Space Shuttle has a quadruplex digital system as its primary and a fifth dissimilar digital channel as a backup.

Heterogeneous Redundancy

As an alternative to identical primaries with dissimilar backups, other architectures use dissimilar primary channels—different hardware, software, and control laws. Dissimilar channels make the voting problem more difficult (how to set thresholds?), but they alleviate problems with generic faults.

This alternative, incidentally, is one in which some of us have placed a lot of trust. It is used on the Airbus A320 transport aircraft, shown in Figure 12. This airplane was certified for commercial service in 1989 and is now flying regular routes in Europe and the United States. Although the hardware architecture of the A320 does not fit exactly into the prototypes we have looked at, it basically consists of four dissimilar channels. Two channels are built out of one brand of microprocessor, and two others are built out of a second brand. The software in each of these similar hardware channels is different, developed by different design teams using different programming languages. Unfortunately, I do not know much about the control laws themselves, but presumably they are similar enough to permit reasonably tight voting thresholds.

In the context of this article, it is important to point out that the A320 is not statically unstable. However, it is a fly-by-wire airplane, which means that there are no mechanical connections between the pilot's input device (a sidestick hand controller, not the traditional control wheel) and the main control surfaces. Only electronic connections exist through the control computers. If these connections fail, the airplane can continue to cruise and can be brought down with the aid of a low-authority mechanical trim system. On the other hand, complete loss of control functions during critical phases of autoland (e.g., just before touchdown) could well be catastrophic.

表决

在这一基本概念过于牢固地扎根于脑海之前，我要提醒大家，图 11 所呈现的示意图隐藏了大多数困难和问题，而正是这些困难和问题使冗余管理成为一门具有挑战性的工程学科。具体而言，图中提到用“某种表决方案”来区分正常通道和故障通道。图中并未展示该方案，但请考虑一下可以如何构建它。基本思路是比较各通道的输出，并按多数通道的结果继续运行。然而，这些输出并非离散的赞成或反对票，而是长度为 8 至 12 位的数字字。在不同通道中，由于传感器读取的信号不同、处理器的运行速率不同，也可能存在其他原因，这些数字字会有所差异。因此，我们不能简单地进行表决，而必须将其与预设阈值进行比较。



图 12. 空中客车 *A320*。

这引出了第一个重大问题：阈值必须有多大？考虑这样一种情况：每个处理器都实现一条包含复杂控制算法的控制律——例如，带有显式辨识的自适应律（如参数估计值按照 $d\,t\,e\phi$ $^T\phi/d$ = +L 演变）。每个通道使用不同的含噪传感器数据执行这条控制律。于是，通道间的差异将由噪声驱动的自由积分构成；根据我们在随机过程入门课程中学到的知识，这些差异表现得像布朗运动，并且会无穷多次地超过任何给定阈值。这对表决可不妙！

请注意，这里的罪魁祸首并不是复杂的控制律。任何具有自由积分或不稳定动态的算法，以及带有模式逻辑、饱和保护和其他离散切换的控制律，都会出现同样的问题。任何输入的微小差异可能导致输出巨大差异的控制算法，都会出现这一问题。

解决这一表决问题的常见方法是利用通道间通信，使所有通道“均衡”（例如，强制它们同步时钟并处理相同的数据）。这样，表决就简化为逐位比较。遗憾的是，这一解决方案只是揭开了困难的第一层，因为使通道均衡所需的通信硬件也必须非常可靠。这需要更多层级的冗余，幸好这种冗余仅涉及较少的组件 [6]。然而，即使通信具有适当的可靠性，仍有某些故障无法被检测到，即那些在每个通道中产生相同症状的故障。

具有相同症状的故障被称为“共性故障”。乍看之下，这类故障似乎不太可能发生（两个或更多相同故障同时发生），但我们随后意识到，硬件或软件中尚未发现的设计错误恰恰具有这一特征。事实上，此类错误存在的可能性不可否认，因此，大多数逐位同步架构也配备了一个或多个备份通道，采用不同的控制律、不同的硬件以及不同的软件。例如，X-29 采用三余度数字系统作为主控制器，同时配备三个备份通道，通过模拟电子电路实现非常基本的控制律。航天飞机采用四余度数字系统作为主系统，并以第五个异构数字通道作为备份。

异构冗余

除了采用相同的主通道配合异构备用通道之外，其他架构还采用异构主通道——使用不同的硬件、软件和控制律。异构通道使表决问题更加困难（如何设定阈值？），但它们能够缓解共性故障带来的问题。

顺便说一下，我们中的一些人对这种替代方案寄予了很大信任。图 12 所示的 Airbus A320 运输机采用了这一方案。该飞机于 1989 年获得商业运营认证，目前正在欧洲和美国的定期航线上飞行。虽然 A320 的硬件架构并不完全符合我们考察过的典型架构，但它基本上由四个异构通道组成。其中两个通道使用一个品牌的微处理器构建，另外两个通道则使用第二个品牌的微处理器构建。这些硬件相似的通道各自采用不同的软件，由不同的设计团队使用不同的编程语言开发。遗憾的是，我对控制律本身了解不多，但想必它们足够相似，能够采用较为严格的表决阈值。

在本文的背景下，有必要指出，A320 并非静不稳定飞机。然而，它是一架电传操纵飞机，这意味着飞行员的输入装置（侧杆手控器，而非传统的驾驶盘）与主操纵面之间没有机械连接。两者之间仅通过飞行控制计算机进行电子连接。如果这些连接失效，飞机仍可继续巡航，并可借助控制权限较低的机械配平系统下降并着陆。另一方面，在自动着陆的关键阶段（例如接地前一刻），控制功能完全丧失很可能造成灾难性后果。

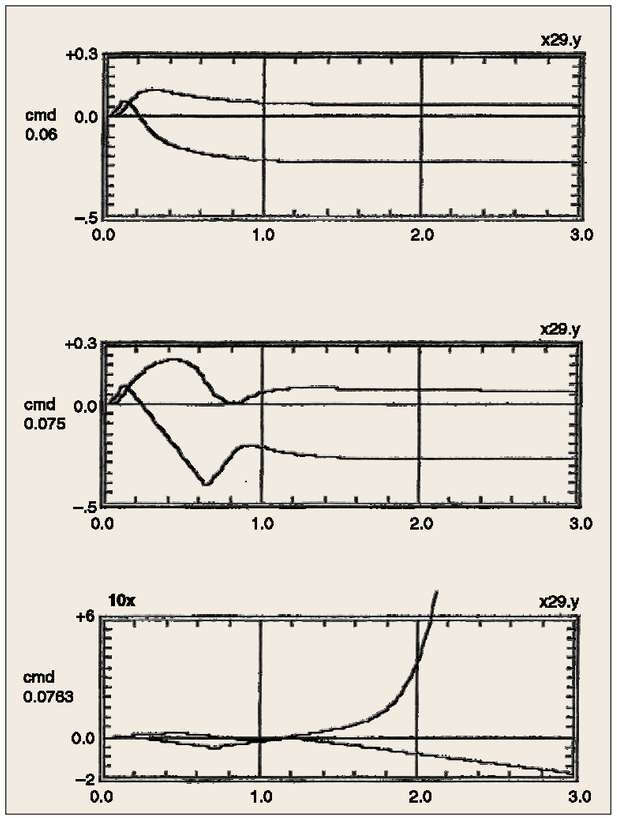


Figure 13. *Simulation traces for the X-29 demonstrating rate limit constraints.*

So in the spirit of my discussions so far, the A320 qualifies as a dangerous system that we should treat with appropriate respect. And the A320 is only the beginning. Both Boeing and McDonnell Douglas expect to build their next transports as fly-by-wire also. It is perhaps comforting to know that neither of these manufacturers expects to build statically unstable commercial transports in the immediate future. Designers at Boeing have told me that unstable fly-by-wire transports are not likely unless they are supersonic. The argument is the same as for the X-29. Relaxed stability for aerodynamic efficiency at supersonic speeds means pronounced instability at subsonic speeds.

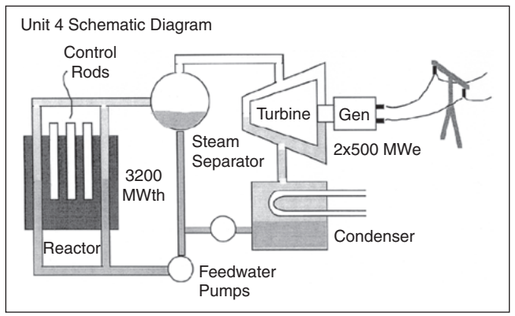


Figure 14. *Schematic diagram of Chernobyl Unit 4 reactor.*

Paper studies of new supersonic transports are ongoing but are not likely to produce commercial vehicles until the next century. But the next century is not far away! Engineers who will design the controls for those vehicles are being trained by us right here today. I, for one, still expect to be traveling when these new vehicles come online, so I have a vested interest that we teach the fundamentals well, not just the formal tools and the mathematics!

Fact 3: Local Stability

Finally, let us turn to the third fact I want to review: Closed-loop systems with unstable components are only locally stable. This should be well known to all of us. An unstable system cannot be stabilized globally with bounded control authority. For linear systems, this follows directly from the equations. Consider motion along an unstable eigenvector. If the control signal is constrained to have a bounded component along that vector, then there exist initial conditions large enough to keep the state-derivative positive, and the system diverges. This argument holds both for magnitude limits on the controls and for rate limits.

An example of local stability due to rate limits is given in Figure 13. This figure shows responses of the prototype X-29 controller from Figure 9 implemented with rate limits and excited by several different step commands. We see that small commands give linear responses. There is about 100% overshoot, consistent with the (unavoidably) large sensitivity of the design. As commands increase, the control rate saturates, and with very little additional command, the system diverges. Please keep the general characteristics of these traces in mind—the triangular waveform of the control at the edge of instability and finally the dramatic divergence with control rate fully saturated. Later I will present some very similar plots, only much more frightening.

For control engineers, local stability means that special care must be taken to avoid excessive commands (i.e., limits on operator inputs). We must also verify that worst-case upsets due to external disturbances do not drive the closed-loop system out of its region of attraction.

These observations bring us back to our first example, the Saab JAS-39 airplane in Figure 1. This airplane was statically unstable, with divergence rates about half as severe as the X-29. Like the X-29, it carried a triplex digital control system with three analog backup channels. As one might ex-

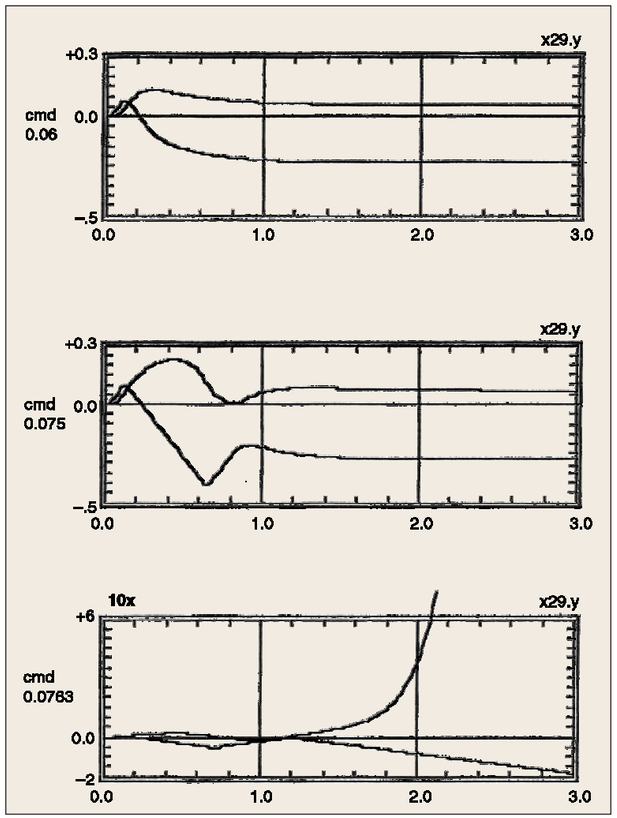


图 13. 展示速率限制约束的 *X-29* 仿真曲线。

因此，按照我此前讨论的精神，A320属于一种危险系统，我们应当给予它应有的尊重。而A320仅仅是个开始。波音和麦道都预计，其下一代运输机也将采用电传操纵。或许令人稍感安心的是，这两家制造商都不打算在不久的未-

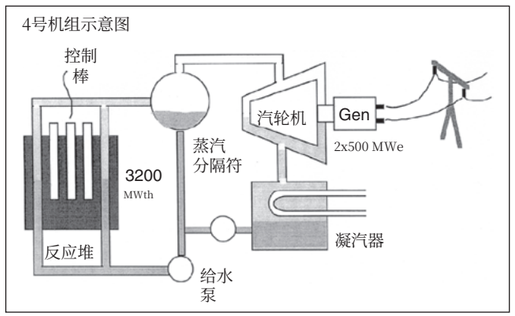


图14. 切尔诺贝利*4*号机组反应堆示意图。

来制造静不稳定的商用运输机。波音的设计人员告诉我，除非是超声速运输机，否则不太可能采用不稳定的电传操纵设计。其理由与X-29相同。为提高超声速飞行时的气动效率而放宽稳定性，意味着在亚声速飞行时会出现显著的不稳定性。

新型超声速运输机的方案研究正在进行，但在下个世纪之前不太可能研制出商用飞机。不过，下个世纪已经不远了！将为这些飞机设计控制系统的工程师，今天就在这里接受我们的培养。就我个人而言，我仍然希望在这些新型飞机投入使用时继续乘机出行，因此，教好基础原理，而不只是形式化工具和数学，与我的切身利益息息相关！

事实 3：局部稳定性

最后，让我们转向我想回顾的第三个事实：含有不稳定部件的闭环系统仅具有局部稳定性。这一点应该是我们所有人都熟知的。控制能力有界时，无法使不稳定系统实现全局稳定。对于线性系统，这可以直接由方程得出。考虑沿不稳定特征向量方向的运动。如果控制信号沿该向量方向的分量被限制为有界，那么就存在足够大的初始条件，使状态导数持续为正，从而导致系统发散。这一论证既适用于控制的幅值限制，也适用于速率限制。

图 13 给出了一个由速率限制导致的局部稳定性示例。该图显示了原型 X-29 控制器的响应，该控制器来自图 9，在实现时加入了速率限制，并由几个不同的阶跃指令激励。可以看到，小幅指令产生线性响应。超调量约为 100 %，这与该设计（不可避免的）高灵敏度一致。随着指令幅值增大，控制速率达到饱和，而指令再增加很少一点，系统就会发散。请记住这些曲线的一般特征——在失稳边缘，控制量呈三角波形；最终，控制速率完全饱和，系统急剧发散。稍后我将展示一些非常相似的图，只是要可怕得多。

对于控制工程师而言，局部稳定性意味着必须特别注意避免过大的指令（即限制操作人员的输入）。我们还必须验证，外部扰动造成的最坏情况偏离不会使闭环系统超出其吸引域。

这些观察结果使我们回到了第一个例子，即图 1 中的 Saab JAS-39 飞机。这架飞机在静态上不稳定，其发散速率约为 X-29 的一半。与 X-29 一样，它配备了三余度数字控制系统，并设有三个模拟备用通道。正如人们可能预

pect, extensive investigations have been going on ever since the airplane's accident, with final official conclusions still not out. However, various reports in the media suggest that the control hardware did not malfunction. Instead, unstable behavior of the control laws in the face of surface rate limits is being cited as the cause. The control laws have already been redesigned to cure the problem (see [7], for example).

Heuristically, we can think of rate limits as acting to reduce available bandwidth for large signal levels (i.e., the controls can only follow position commands whose frequency-amplitude product is less than the rate limit). This suggests that the effects of rate limits can indeed be alleviated by redesigning control laws for lower assumed available bandwidth. According to Bode's integrals, however, such redesigned controllers will achieve poorer sensitivity performance, especially for unstable problems. So the improvements that can be gained for the JAS-39 will necessarily be limited.

Once again, it is evident that unstable/dangerous systems must be treated with more than casual respect.

The Chernobyl Story

Finally, to make this point most dramatically, let me recount the story of the nuclear accident at Chernobyl (Figure 2). On 28 April 1986, news came out of Ukraine that a nuclear power plant had destroyed itself two days earlier and had released significant amounts of radioactive contaminants over a wide area. Short of nuclear war or impending long-term climate changes, this kind of accident certainly looms large in the public mind as one of the more serious threats to our well being.

Whether we choose to recognize it or not, control played a major role in that accident. The plant's hardware did not fail. No valve hung up, no electronic box went dead, and no metallurgical flaw caused a critical part to break. Instead, the reactor control system systematically drove the plant into an operating condition from which there was no safe way to recover. This is true, at least, if we count the control system's hardware, its human operators, and its operating policies as part of the system.

The Plant

To tell this story, we need a few facts about the plant itself. The information comes from a seminar given by Herbert Kouts of Brookhaven National Laboratory summarizing the accident [8]. I want to emphasize, however, that the very simplified control interpretations I am about to make are my own, so the blame for any errors and absurdities is mine as well.

The plant at Chernobyl consisted of four units, each laid out as shown in Figure 14. Unit 4 is the one that experienced the accident. It had a boiling water reactor that took in water at the bottom and heated it to produce a mixture of water and steam. The steam was separated in steam drums and drove conventional turbine-generators to supply power to the distribution grid. There were two turbines rated at 500 MWe (electrical) each and two complete water/steam flow circuits through the reactor. The water/steam circuits operated at about 1,000 psi pressure, with a boiling temperature inside the reactor of about 540 °F.

Chernobyl
Events of the Night of 25 April 1986

- 1) Power had been brought down during the previous day to around 700 MWt, the edge of the legal low-power operating limit, ready to run the test.
- 2) A switchover was made to different flux detectors, better suited for power sensing at this low level, which was apparently a standard procedure. During the switchover, however, the operator neglected to engage the power-hold mode. This oversight set up the conditions for the accident.
- 3) Without automatic power hold, reactor power dropped rapidly to 30 MWt. The operator halted this drop and recovered to 200 MWt by withdrawing control rods.
- 4) With automatic control of power and manual control of feedwater, the plant successfully maintained 200 MWt. Because feedwater flow settings were high, however, the steam void in the reactor dropped to zero. Lower reactivity was compensated for by pulling even more control rods. Only six to eight rods remained in the reactor, far fewer than the minimum number (30) required by operating regulations.
- 5) To avoid automatic shutdown triggered by out-of-range steam drum and feedwater signals, the operator disabled the associated automatic scram control circuits.
- 6) Recognizing excessive feedwater, the operator finally reduced pumping rates. The steam void recovered, producing increased reactivity. Automatic power controls responded to keep power regulated. This response was rate limited and barely stable.
- 7) Next, in preparation for the actual intended test, the operator disabled the automatic scram circuits associated with turbine trip signals.
- 8) Finally, the test was actually started. Steam was cut to the test turbine. The steam void began to rise, and the power controller responded by inserting all three available banks of control rods at maximum rates. This was too little control authority, applied too slowly. A huge power rise followed, up to an estimated 300,000 MWt (100 times rated capacity). The reactor was destroyed. Steam at primary working pressure was released into the reactor containment chamber. The 1,000-ton cover plate of the chamber blew off, and the entire radioactive debris was exposed to the environment.

料的那样，自飞机发生事故以来，广泛的调查一直在进行，最终的官方结论仍未公布。然而，媒体的各种报道表明，控制硬件并未发生故障。相反，控制律在舵面速率限制下表现出的不稳定行为被认为是事故原因。控制律已经重新设计，以解决这一问题（see [7], for 示例）。

直观地说，我们可以认为速率限制会降低大信号水平下的可用带宽（即，控制装置只能跟随频率与幅值的乘积小于速率限制的位置指令）。这表明，通过假定较低的可用带宽来重新设计控制律，确实可以减轻速率限制的影响。然而，根据 Bode 积分，这样重新设计的控制器在灵敏度方面的性能会更差，尤其是对于不稳定问题。因此，JAS-39 所能获得的改进必然是有限的。

这再次清楚地表明，对于不稳定/危险的系统，必须给予充分的重视，绝不能掉以轻心。

切尔诺贝利的故事

最后，为了最鲜明地说明这一点，让我讲述一下切尔诺贝利核事故的经过（图 2）。1986 年 4 月 28 日，乌克兰传出消息，一座核电站在两天前自毁，并向广大区域释放了大量放射性污染物。除了核战争或即将到来的长期气候变化，这类事故无疑是公众心目中对我们福祉较为严重的威胁之一。

无论我们是否愿意承认，控制在那次事故中都起了重大作用。核电站的硬件并未发生故障。没有阀门卡住，没有电子设备失灵，也没有冶金缺陷导致关键部件断裂。相反，反应堆控制系统一步步将核电站推入了一种无法安全恢复的运行状态。至少，如果我们将控制系统的硬件、操作人员及其运行规程都视为系统的一部分，这一说法就是成立的。

被控对象

要讲述这个故事，我们需要了解一些关于核电站本身的事实。这些信息来自 Brookhaven 国家实验室的 Herbert Kouts 所作的一场研讨会报告，报告概述了 accident [8]. 然而，我想强调，我接下来将从控制角度作出的高度简化的解释是我个人的看法，因此，任何错误和荒谬之处也都由我负责。

切尔诺贝利核电站由四个机组组成，每个机组的布置均如图 14 所示。发生事故的是 4 号机组。该机组采用沸水反应堆，水从底部进入，经加热后形成水和蒸汽的混合物。蒸汽在汽包中分离，并驱动常规汽轮发电机向配电网供电。该机组有两台额定电功率各为 500 MWe 的汽轮机，以及两套完整的流经反应堆的水/蒸汽回路。水/蒸汽回路的运行压力约为 1,000 psi，反应堆内的沸腾温度约为 540 °F。

切尔诺贝利
1986 年 4 月 25 日晚发生的事件

• 1) 在前一天，功率已降至约 700 MWt，即法规规定的低功率运行限值的边界，准备进行试验。• 2) 随后切换到另一组更适合在这一低功率水平下测量功率的中子通量探测器，这显然是一项标准操作程序。然而，在切换过程中，操作员忘记启用功率保持模式。这一疏忽为事故的发生创造了条件。• 3) 由于没有自动功率保持功能，反应堆功率迅速降至 30 MWt。操作员通过抽出控制棒制止了功率下降，并使功率恢复至 200 MWt。• 4) 在自动控制功率、手动控制给水的情况下，机组成功地将功率维持在 200 MWt。然而，由于给水流量设定值偏高，反应堆内的蒸汽空泡降至零。为补偿反应性的降低，操作员抽出了更多控制棒。反应堆内仅剩六至八根控制棒，远少于运行规程要求的最少数量 (30)。• 5) 为避免汽包和给水信号超出范围而触发自动停堆，操作员停用了相关的自动紧急停堆控制电路。• 6) 操作员意识到给水过量，终于降低了泵送流量。蒸汽空泡恢复，导致反应性增加。自动功率控制系统作出响应，以维持功率调节。该响应受到速率限制，仅勉强保持稳定。• 7) 接下来，为准备实际计划进行的试验，操作员停用了与汽轮机跳闸信号相关的自动紧急停堆电路。• 8) 最后，试验正式开始。通往试验汽轮机的蒸汽被切断。蒸汽空泡开始增加，功率控制器作出响应，以最大速率插入全部三组可用控制棒。控制能力太小，施加速度也太慢。随后功率急剧上升，估计高达 300,000 MWt（额定容量的 100 倍）。反应堆被毁。处于一回路工作压力下的蒸汽释放到反应堆安全壳腔室中。腔室重达 1,000 吨的盖板被掀飞，全部放射性碎片暴露于环境中。

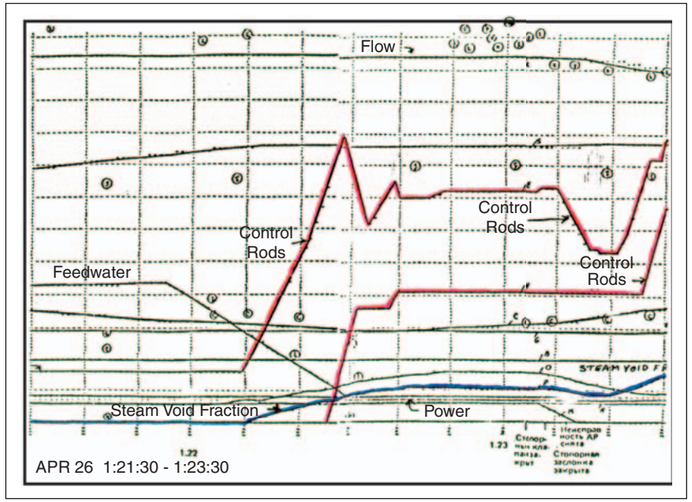


Figure 15. *Trace of reactor events preceding Chernobyl disaster.*

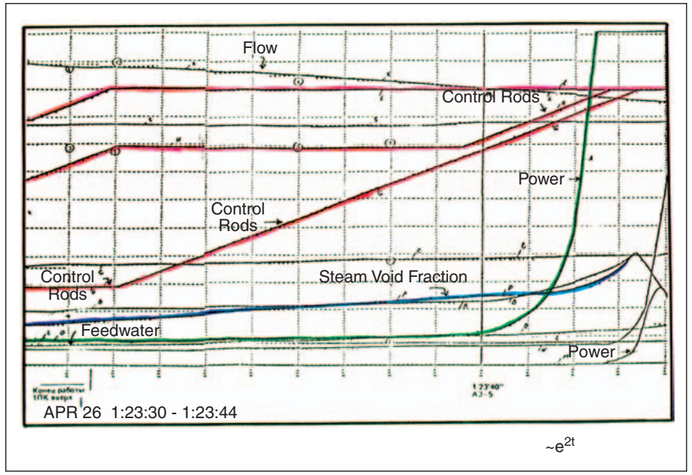


Figure 16. *Trace of power increase associated with the Chernobyl disaster.*

The reactor itself consisted of a graphite core with many pressure tubes (≈1,600) through which the water flowed to be boiled. Each tube contained several fuel rods of uranium oxide that produced the nuclear fission reaction. The generated thermal power was measured by ion flux devices and controlled by several banks of control rods, inserted or withdrawn from the core to moderate the reaction. A second major control input was provided by the feedwater pumps, which regulated flow rates into the bottom of the reactor.

The reason this plant is of interest is that its reactor design has a so-called positive void coefficient. The void refers to the ratio of steam to water in the pressure tubes. The void coefficient refers to the gradient of thermal power with respect to the void. In Chernobyl's design, this coefficient is positive at low power, leading in the linearized sense to a first-order, open-loop instability. Heuristically, as more steam fills the tubes, the reaction becomes stronger (water is apparently a better moderator than steam). This produces more steam, which produces more reactivity, and so it goes.

At nominal operating power, 3,200 MWt (thermal), the positive void coefficient is swamped by other effects and the reactor design is stable. For this reason, operating policies only permitted sustained operation above a minimum power threshold of 700 MWt. Of course, to get the reactor started and stopped, it had to transition through the unstable region.

An Experiment

The accident occurred in connection with an experiment conducted on the electrical side of the plant. The operators wanted to evaluate a scheme for drawing electrical power from a turbine-generator coasting down after a trip from the main power grid. During such a trip, electrical power at the plant is momentarily lost, the reactor scrams (shuts down) automatically, and backup diesels are started to resupply local electrical equipment at the plant. The new scheme would go through this sequence without local power interruption.

The test was scheduled in conjunction with a routine reactor shutdown for maintenance. The intent was to slowly bring power down into the 700- to 1,000-MWt range, load the test turbine-generator with some of the circulation pumps, cut steam to the turbine, and scram the reactor.

A summary of the events of that evening is given in the sidebar. Traces of the last few seconds of this event are shown at the tail end of Figure 15 and continue on an expanded time scale into Figure 16. We can see a slight steam void drop as pressure increased momentarily after steam was cut to the turbine. The steam void then began to build steadily, the control rods dropped in at maximum rate, and finally the power rose uncontrollably. Simple time-to-double calculations approximated from the final power rise place the reactor's unstable pole between 1.5

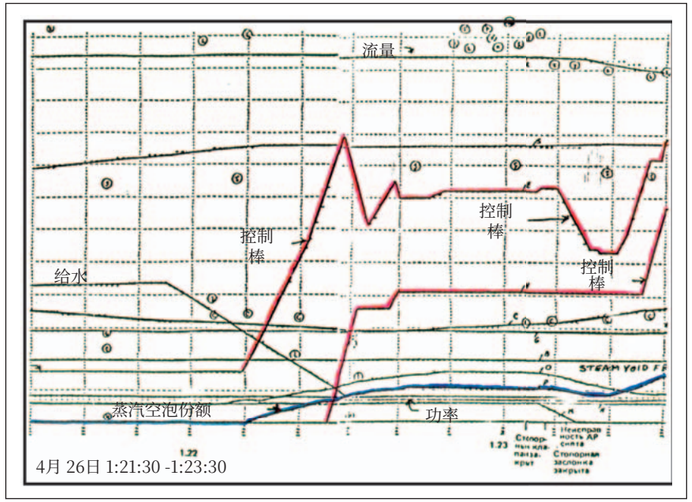


图 15. 切尔诺贝利灾难发生前的反应堆事件轨迹。

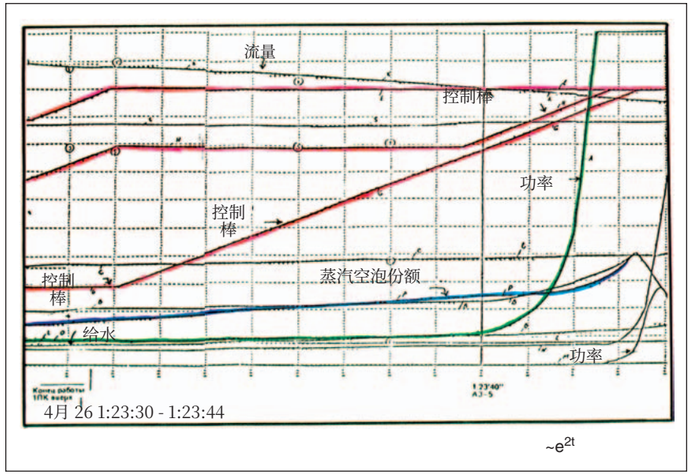


图 16. 与切尔诺贝利灾难相关的功率上升曲线。

反应堆本体由石墨堆芯和许多压力管（≈1,600）组成，水流经这些压力管并被加热至沸腾。每根管内装有数根氧化铀燃料棒，用于产生核裂变反应。产生的热功率由离子通量装置测量，并由数组控制棒控制；通过将控制棒插入堆芯或从堆芯中抽出，调节反应。第二个主要控制输入由给水泵提供，用于调节进入反应堆底部的流量。

这座核电站之所以值得关注，是因为其反应堆设计具有所谓的正空泡系数。空泡是指压力管中蒸汽与水的比例。空泡系数是指热功率相对于空泡的梯度。在切尔诺贝利的设计中，该系数在低功率下为正，从线性化的角度来看，这会导致一阶开环不稳定性。直观地说，随着更多蒸汽充满管道，反应会变得更强（水显然比蒸汽更适合作为慢化剂）。这会产生更多蒸汽，进而产生更高的反应性，如此循环下去。

在额定运行功率 3,200 MWt（热功率）下，正空泡系数的影响被其他效应所压倒，反应堆设计是稳定的。因此，运行规程仅允许在高于 700 MWt 的最低功率阈值时持续运行。当然，为了启动和停止反应堆，必须经过不稳定区域。

一项试验

事故发生时，电厂正在进行一项电气系统试验。操作人员希望评估一种方案，即在汽轮发电机组与主电网跳闸解列后，利用其惰转过程提供电力。在这种跳闸过程中，电厂会短暂失电，反应堆自动紧急停堆（关闭），备用柴油发电机启动，为电厂内的电气设备恢复供电。新方案将使这一过程在厂内供电不中断的情况下完成。

该试验计划结合反应堆例行停堆检修进行。计划是缓慢地将功率降至 700- 至 1,000-MWt 范围内，并将部分负荷接至试验用汽轮发电机组，这些负荷属于循环泵，切断通往汽轮机的蒸汽，并使反应堆紧急停堆。

侧栏概述了当晚发生的事件。该事件最后几秒的曲线显示在图 15 的末尾，并在图 16 中以展开的时间尺度继续呈现。我们可以看到，在切断通往汽轮机的蒸汽后，压力瞬间上升，蒸汽空泡略有减少。随后，蒸汽空泡开始持续增加，控制棒以最大速率插入，最终功率失控上升。根据最终的功率上升过程近似计算的简单倍增时间表明，反应堆的不稳定极点位于 1.5

and 2.0 rad/s, equivalent to a 10-ft broomstick. Again, except for signs, these traces are very similar to Figure 13.

Back to the Punch Line

Why have I taken the time to describe these details of the Chernobyl accident? Certainly I do not want to focus on its tragic consequences, nor do I want to make a case for antinuclear advocates. I simply find this accident to be the most compelling example of blatant disregard for the basic facts I have restated here. We all claim to know these facts, to respect them, and to teach them. Yet the operators at Chernobyl did not appear to know them. Indeed, it can be argued that the plant's designers did not know them either. How else would they be content to write paper regulations prohibiting operation at low power and not insist on reliable hardware to enforce such regulations? On at least two occasions, the operators were able simply to shut off critical safety circuits. Had these shutoffs not been made or not been possible, there would have been adequate time to scram the reactor automatically and to save the plant, even after item 6 of the fateful sequence of events. In redundancy management terms, this possibility qualifies as a generic fault, a single design flaw able to defeat the safety of the entire control architecture.

The Chernobyl accident also provides the most compelling motivation I can think of for us to improve the way we do our job. This reactor control application, as well as the airplane applications I talked about earlier, illustrate that society does indeed permit control engineers to operate dangerous systems. The number of such applications increases steadily. Not all of them have such dramatic consequences as Chernobyl, but they are dangerous nevertheless. Control designers and operators appreciate and respect the practical, physical consequences of these applications only to the extent that we, their teachers, value and instill that appreciation. Unfortunately, our behavior over the past few years at conferences, in our journals, and I suspect also in our lecture halls, places little value upon it. Instead, our behavior tends to uphold mathematical rigor as the only virtue to strive for in control. This trend is incompatible with the trend in applications.

We must place renewed emphasis on stating and teaching the principles of our subject clearly and well. The applications out there are simply too serious for us to hide from responsibility under a cloak of mathematics.

Historical Notes

Much has changed since this article was first presented as the Bode Prize Lecture in December 1989. Here are some brief notes of how certain topics in the lecture have played out:

1) The SAAB JAS-39 accident was indeed attributed to unstable oscillations involving actuator saturations. Control laws were redesigned and retested. The new design experienced a second accident in August 1993, for similar causes. Another redesign followed, this time successful, and the aircraft reached a production milestone with the delivery of 30 aircraft completed in 1996. A prototype of the USAF F-22 fighter also experienced unstable oscillations in 1992, barely avoiding loss of the aircraft.
2) The X-29 research aircraft completed its flight test program without incident. It was retired in 1992 after 374 test flights.
3) After a shaky start, the Airbus A320 fly-by-wire commercial transport has accumulated a strong record of safety and performance and continues to be in service worldwide. It has been joined by Boeing's 777, also fly-by-wire, with a similar strong safety and performance record. No supersonic transports are foreseen in the near future. Design studies during the late 1990s showed their economics to be unattractive using currently projected material and propulsion technologies.
4) The sarcophagus at Chernobyl continues to stand as a stark reminder of the need to "respect the unstable."

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至 2.0rad/s 之间，相当于一根 10-ft 长的扫帚柄。同样，除符号外，这些曲线与图 13 非常相似。

回到核心论点

我为什么要花时间描述切尔诺贝利事故的这些细节？当然，我不想把重点放在其悲惨后果上，也不想为反核倡导者提供论据。我只是认为，这起事故是公然无视我在此重申的基本事实的最有说服力的例子。我们都声称了解这些事实、尊重这些事实，并在教学中传授这些事实。然而，切尔诺贝利的操作人员似乎并不了解这些事实。事实上，可以说，核电站的设计者也不了解这些事实。否则，他们怎么会满足于制定禁止低功率运行的纸面规定，却不坚持采用可靠的硬件来强制执行这些规定？至少有两次，操作人员轻易就关闭了关键安全电路。如果他们没有关闭这些电路，或者根本无法关闭这些电路，那么即使在那一连串致命事件的第 6 项发生之后，仍有足够的时间使反应堆自动紧急停堆并挽救核电站。用冗余管理的术语来说，这种可能性属于一种共性故障，即一个足以使整个控制架构的安全保障失效的设计缺陷。

切尔诺贝利事故也为我们改进工作方式提供了我能想到的最有说服力的动力。这一反应堆控制应用，以及我前面谈到的飞机应用，都表明社会确实允许控制工程师操作危险系统。此类应用的数量正在持续增加。并非所有应用都会产生像切尔诺贝利事故那样严重的后果，但它们仍然具有危险性。控制系统设计者和操作人员能在多大程度上认识并尊重这些应用在实际中产生的物理后果，完全取决于我们这些教师有多重视这种认识，以及有多努力地将其传授给他们。遗憾的是，过去几年里，我们在会议上、期刊中，以及我怀疑也包括课堂上的表现，都很少体现出对这种认识的重视。相反，我们的表现倾向于将数学严谨性奉为控制领域唯一值得追求的美德。这一趋势与应用领域的发展趋势不相容。

我们必须重新重视清晰、透彻地阐述和讲授本学科的原理。现实中的这些应用事关重大，我们不能躲在数学的外衣下逃避责任。

历史回顾

自本文于1989年12月首次作为博德奖讲座发表以来，许多情况发生了变化。以下简要说明讲座中某些主题的后续发展：

1) SAAB JAS-39 事故确实被归因于涉及执行器饱和的不稳定振荡。控制律经过了重新设计和重新测试。新的设计在 August 1993 因类似原因遭遇了第二次事故。随后又进行了重新设计，这次取得了成功，飞机也达到了一个生产里程碑：在 1996 完成了 30 架飞机的交付。美国空军 F-22 战斗机的一架原型机也在 1992 出现了不稳定振荡，险些损失该机。

2) X-29研究机顺利完成了飞行试验计划，期间未发生事故。在完成374次试飞后，该机于1992年退役。

3) 空客A320电传操纵民用运输机在经历了一个不太顺利的开端后，积累了良好的安全与性能记录，并继续在世界各地服役。波音的777也加入了这一行列，同样采用电传操纵，也取得了类似的良好安全与性能记录。预计近期内不会出现超声速运输机。1990年代末的设计研究表明，采用目前所预计的材料和推进技术，其经济性缺乏吸引力。
4) 切尔诺贝利的石棺仍然矗立，鲜明地提醒人们必须“尊重不稳定性”。

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